

Explanatory PMHD

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2026-04-13

```
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.5.2
## v ggplot2     4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr       1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(ggplot2)
library(lattice)
library(viridis)

## Warning: package 'viridis' was built under R version 4.5.3
## Loading required package: viridisLite

main.data <- read.csv("D:/UHasselt/2nd semester/PMHD/project/G12.outcome.data.csv",
                      sep = ",")
str(main.data)

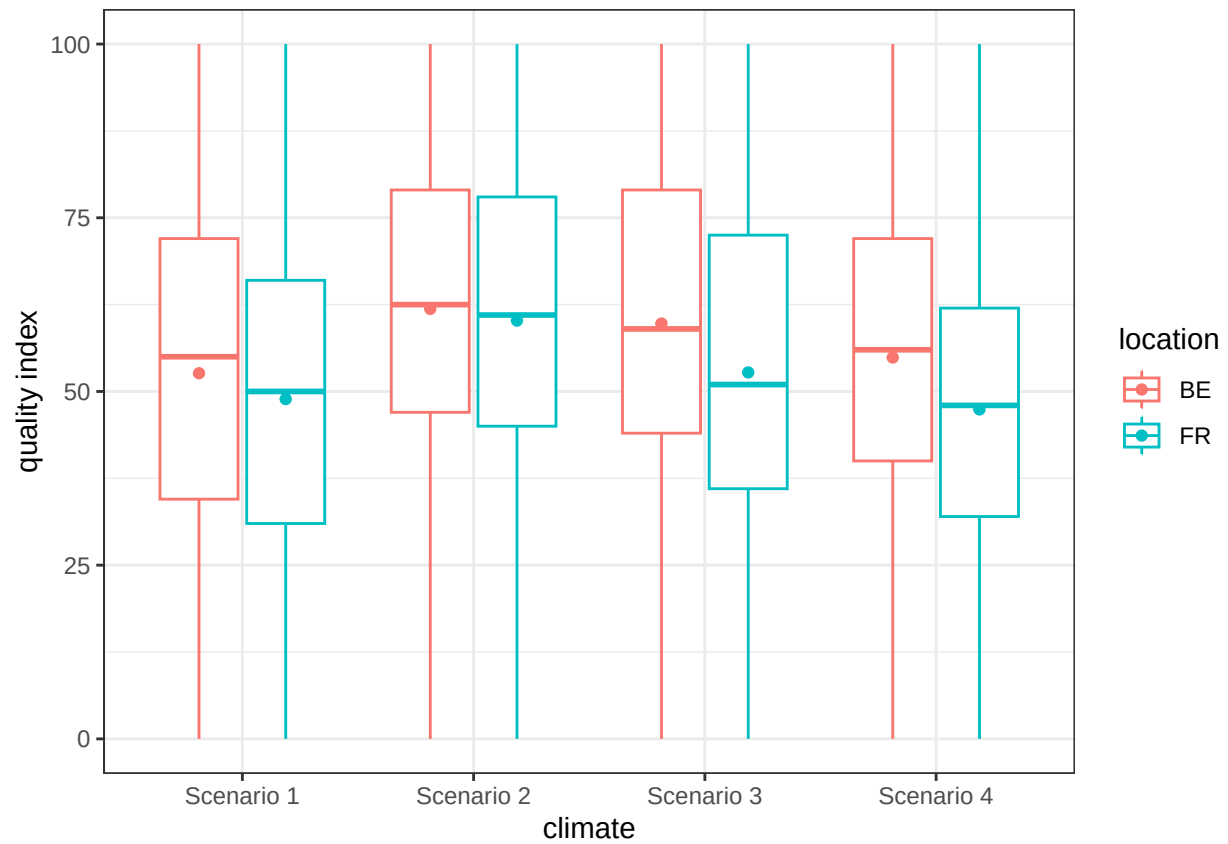
## 'data.frame':    1942 obs. of  8 variables:
## $ ID           : chr  "BE_1_1_1" "BE_1_1_2" "BE_1_1_3" "BE_1_1_4" ...
## $ location      : chr  "BE" "BE" "BE" "BE" ...
## $ ecotr_id      : int   1 1 1 1 1 1 1 1 1 1 ...
## $ tree_id       : int   1 1 1 1 1 1 1 1 1 1 ...
## $ pear_id       : int   1 2 3 4 5 6 7 8 9 10 ...
## $ species       : chr   "Conference" "Conference" "Conference" "Conference" ...
## $ climate       : chr   "Scenario 1" "Scenario 1" "Scenario 1" "Scenario 1" ...
## $ quality_idx   : int   82 39 46 42 90 44 14 27 98 38 ...

sapply(main.data, function(x) sum(is.na(x)))

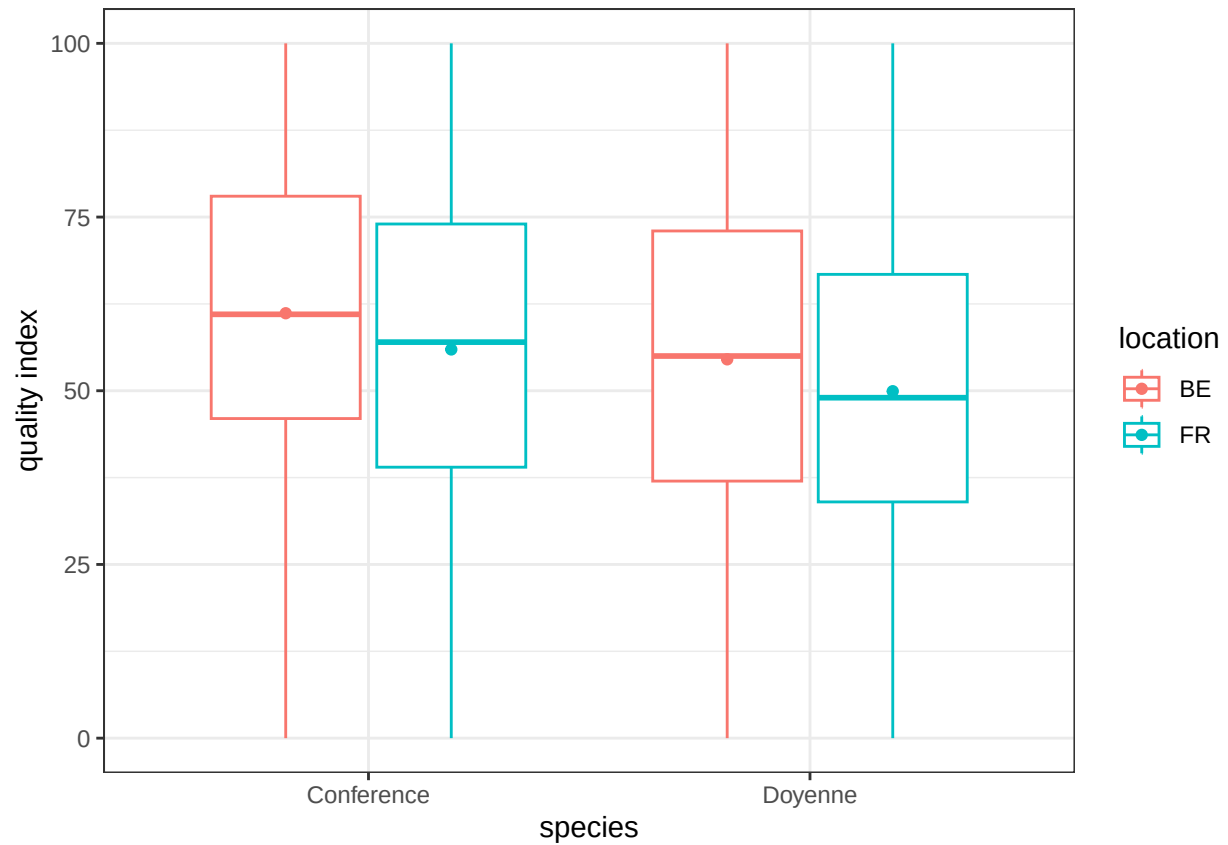
##           ID      location      ecotr_id      tree_id      pear_id      species
##           0           0           0           0           0           0
## climate quality_idx
##           0           0

ggplot(main.data, aes(climate, quality_idx, colour = location))+
  geom_boxplot()+
  stat_summary(fun = mean, geom = "point",
              position = position_dodge(width = 0.75)) +
```

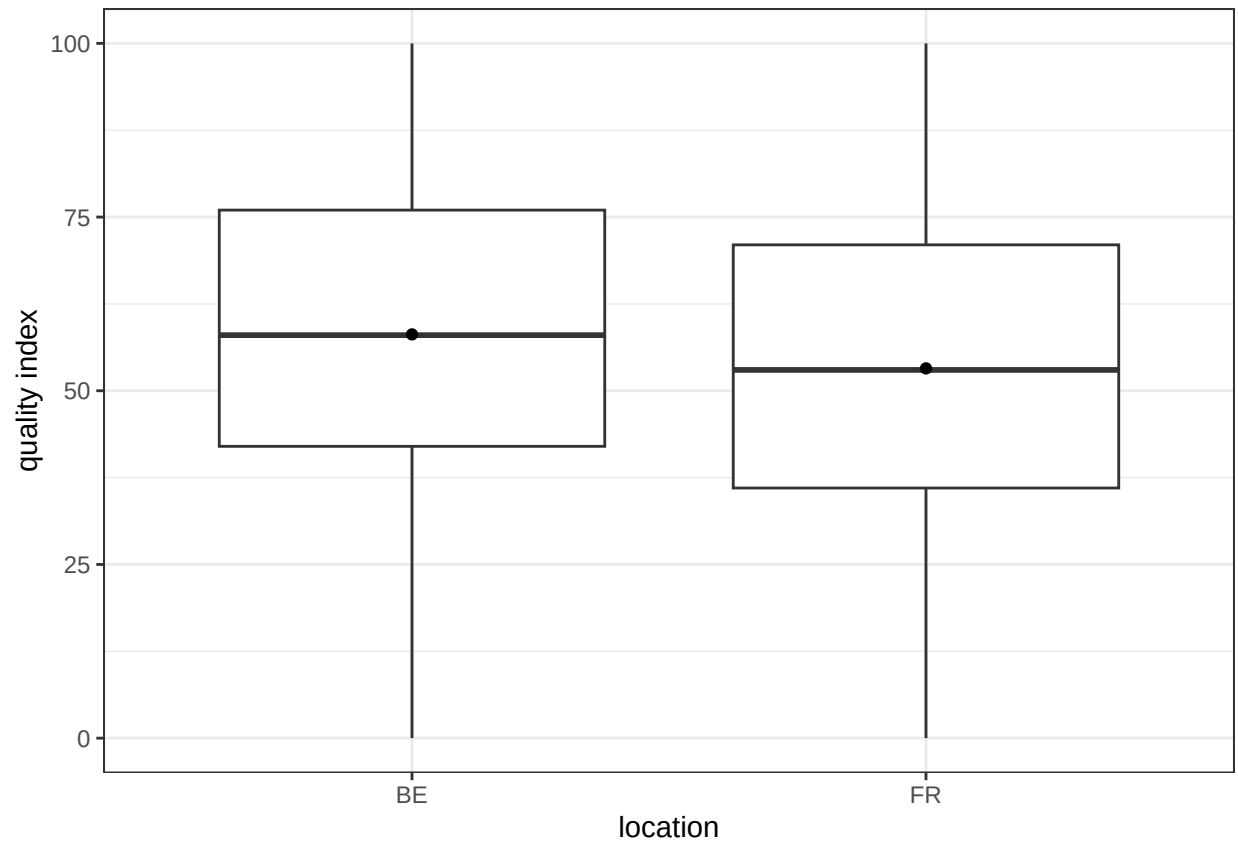
```
labs(y="quality index")+
theme_bw()
```



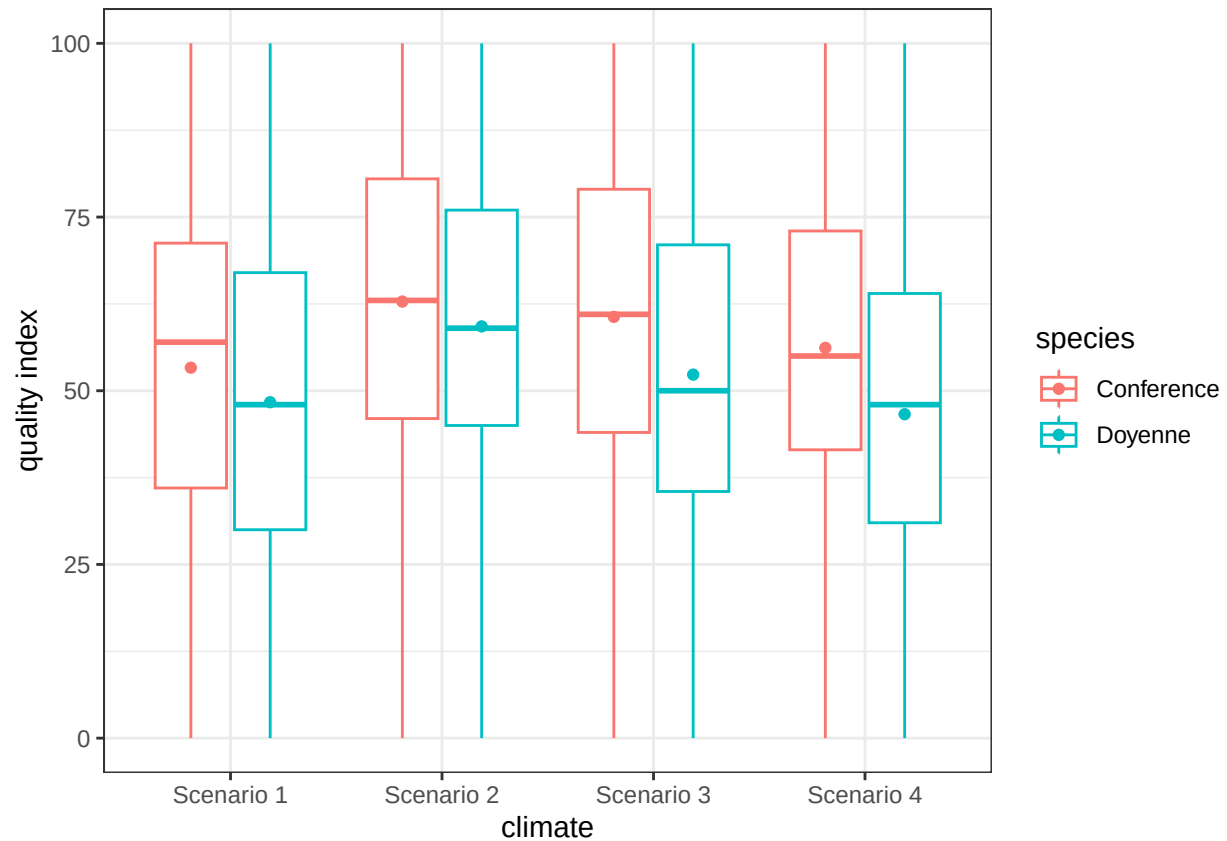
```
ggplot(main.data, aes(species, quality_idx, colour = location))+
  geom_boxplot()+
  stat_summary(fun = mean, geom = "point",
               position = position_dodge(width = 0.75)) +
  labs(y="quality index")+
  theme_bw()
```



```
ggplot(main.data, aes(location, quality_idx)) +  
  geom_boxplot() +  
  stat_summary(fun = mean, geom = "point",  
               position = position_dodge(width = 0.75)) +  
  labs(y="quality index") +  
  theme_bw()
```

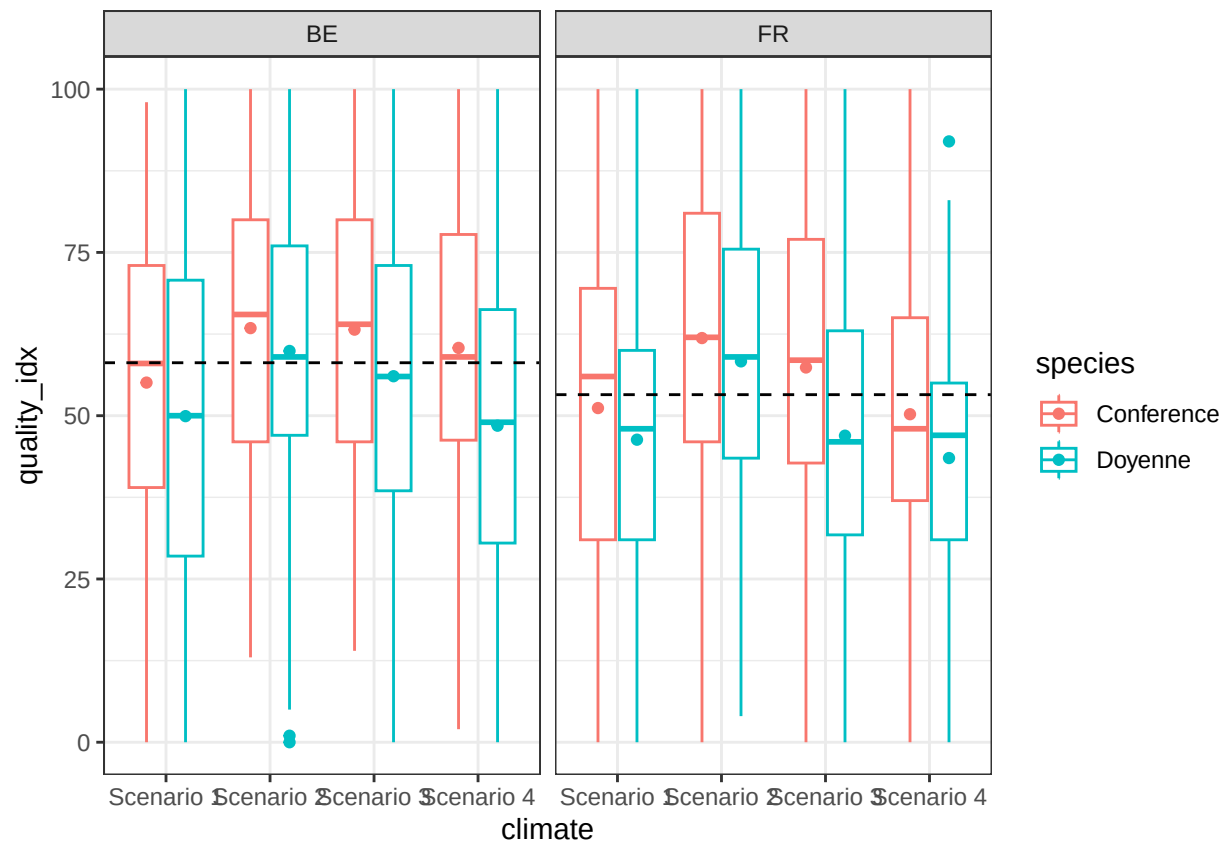


```
ggplot(main.data, aes(climate, quality_idx, colour = species)) +  
  geom_boxplot() +  
  stat_summary(fun = mean, geom = "point",  
               position = position_dodge(width = 0.75)) +  
  labs(y="quality index") +  
  theme_bw()
```

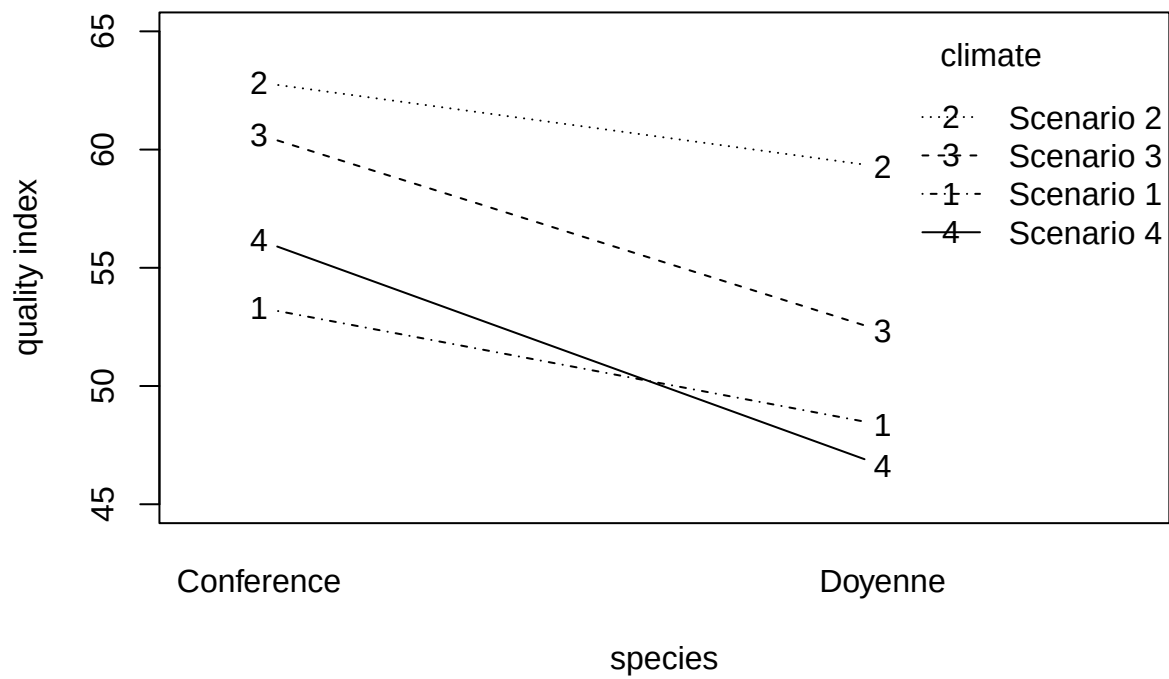


```
mean_data <- main.data %>%
  group_by(location) %>%
  summarise(mean_quality = mean(quality_idx))

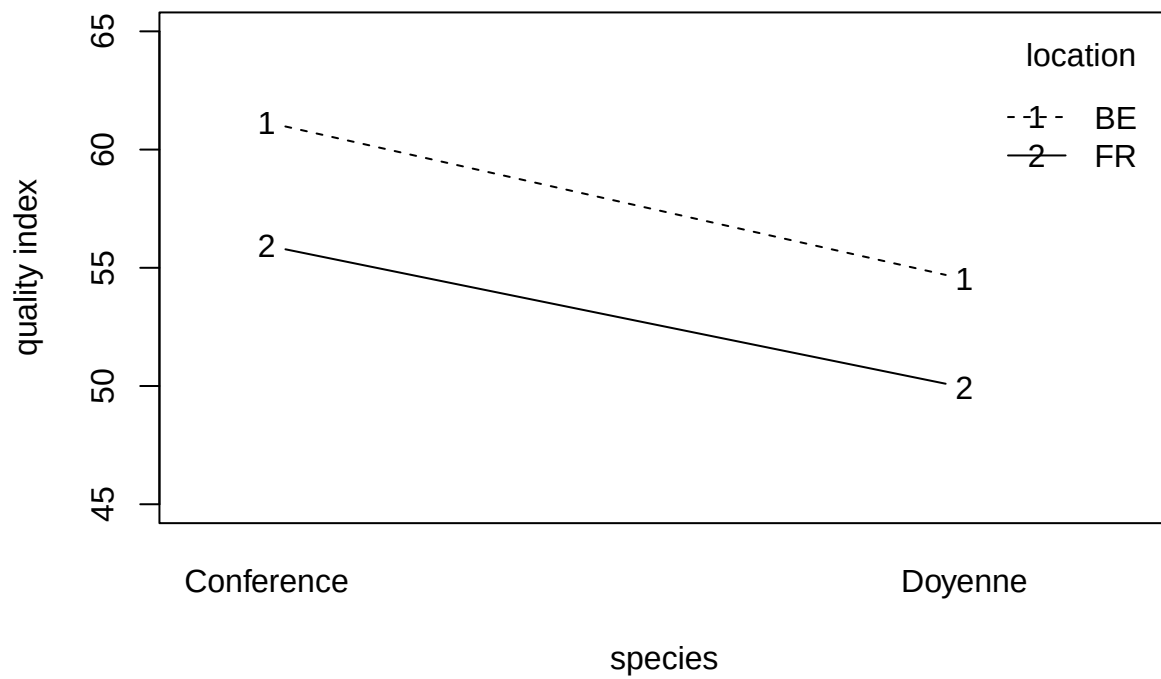
ggplot(main.data, aes(climate, quality_idx, colour = species)) +
  geom_boxplot() +
  stat_summary(fun = mean, geom = "point",
              position = position_dodge(width = 0.75)) +
  geom_hline(data = mean_data,
            aes(yintercept = mean_quality),
            linetype = "dashed",
            color = "black") +
  facet_wrap(~location) +
  theme_bw()
```



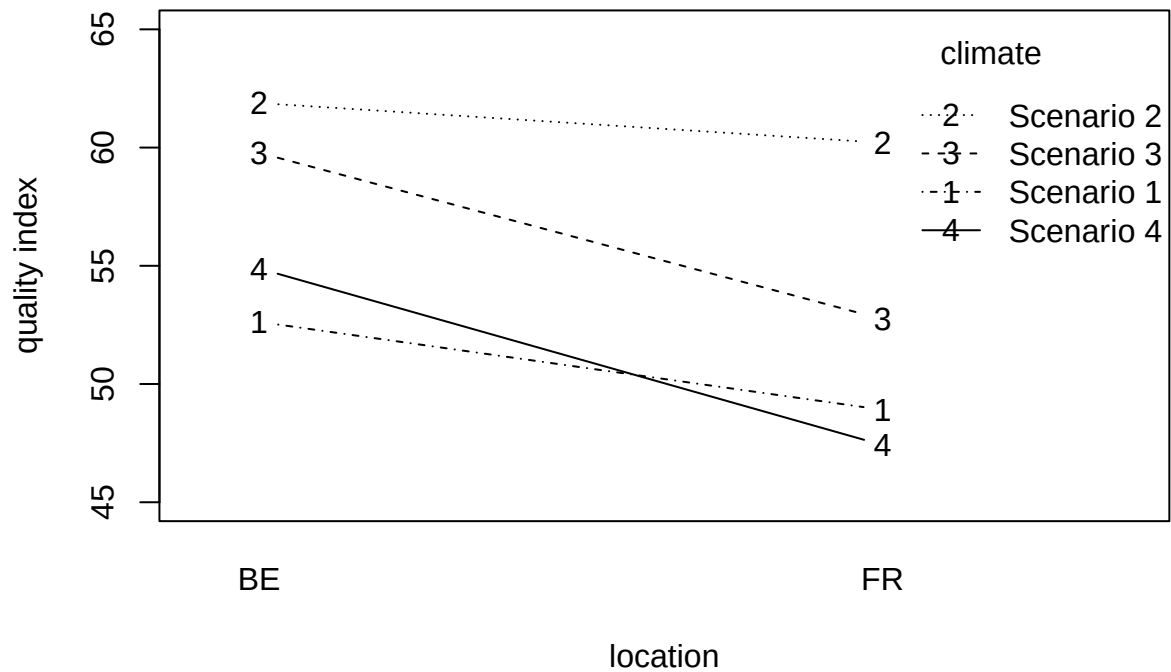
```
interaction.plot(x.factor = main.data$species,
  trace.factor = main.data$climate,
  response = main.data$quality_idx,
  trace.label = "climate",
  type = "b",
  xlab = "species",
  ylab = "quality index",
  ylim = c(45, 65))
```



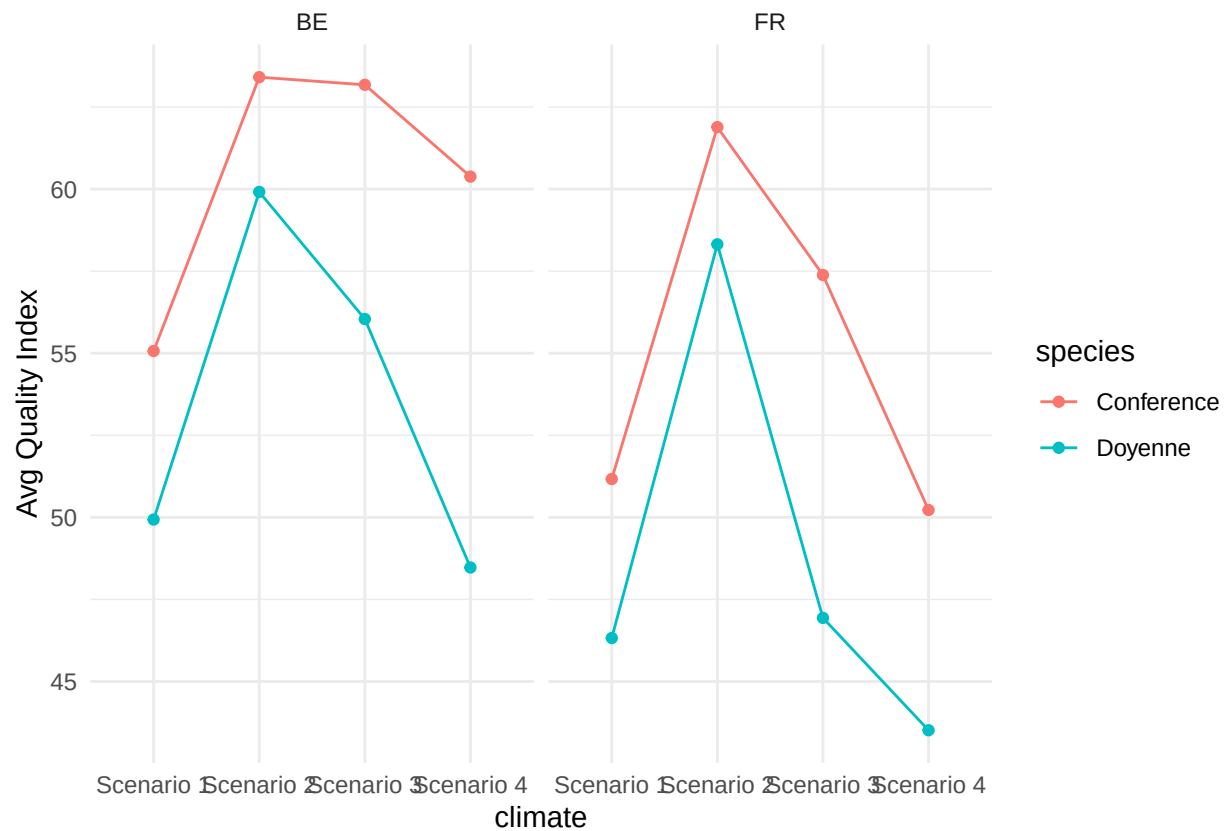
```
interaction.plot(x.factor = main.data$species,
  trace.factor = main.data$location,
  response = main.data$quality_idx,
  trace.label = "location",
  type = "b",
  xlab = "species",
  ylab = "quality index",
  ylim = c(45, 65))
```



```
interaction.plot(x.factor = main.data$location,  
  trace.factor = main.data$climate,  
  response = main.data$quality_idx,  
  trace.label = "climate",  
  type = "b",  
  xlab = "location",  
  ylab = "quality index",  
  ylim = c(45, 65))
```

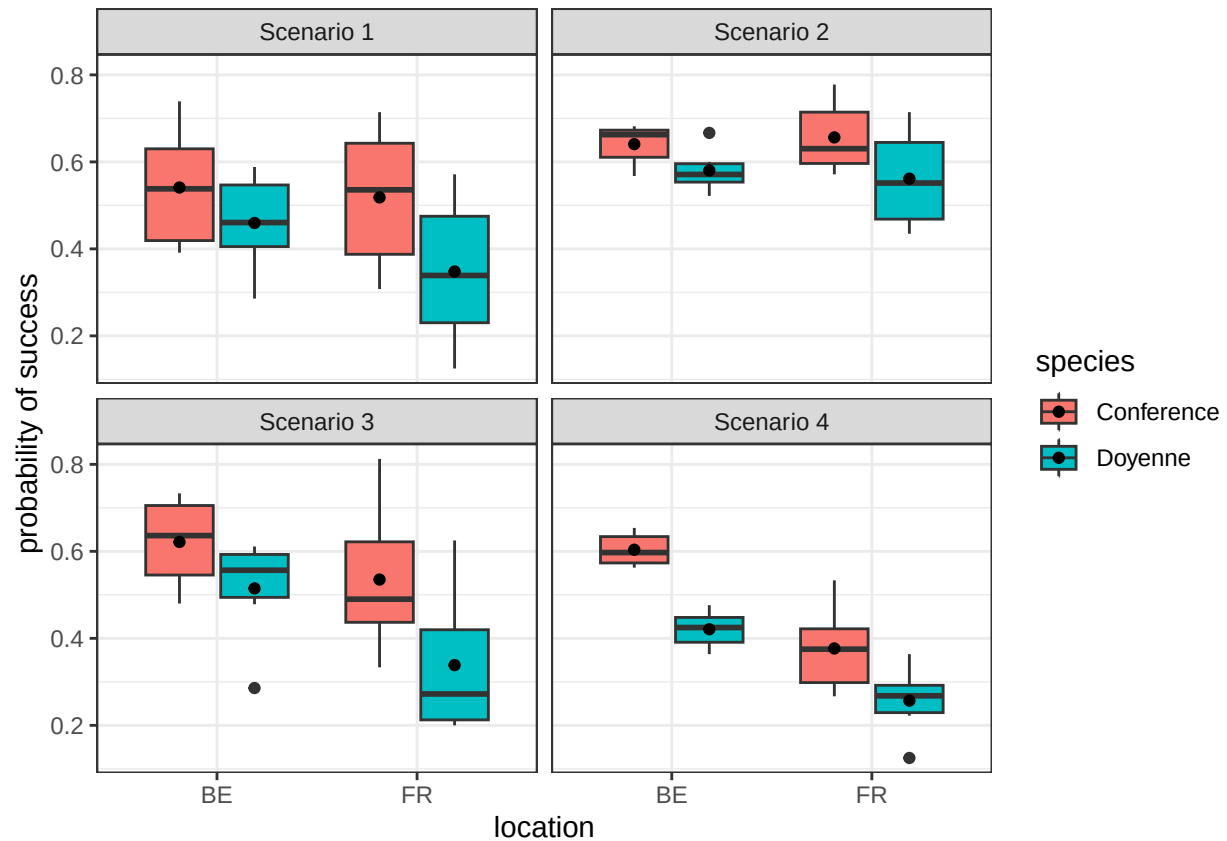
```
ggplot(main.data, aes(x = climate, y = quality_idx, color = species, group = species)) +
  stat_summary(fun = mean, geom = "line") +
  stat_summary(fun = mean, geom = "point") +
  facet_grid(~location) +
  theme_minimal() +
  labs(y = "Avg Quality Index")
```



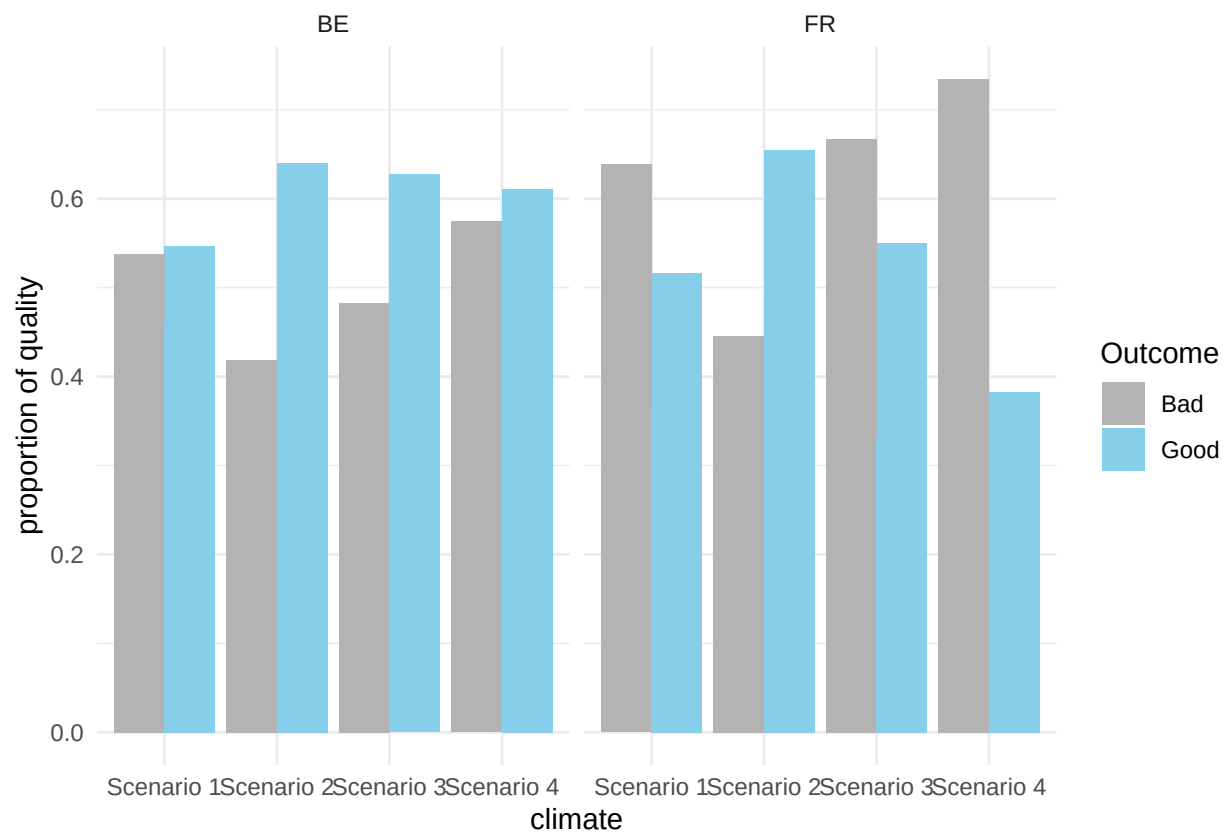
```
main.data2 <- main.data %>%
  mutate(quality_b = if_else(quality_idx >= 55,1,0))

# write.csv(main.data2,"D:/UHasselt/2nd semester/PMHD/project/quality_binary.csv",
#           row.names = FALSE)
```

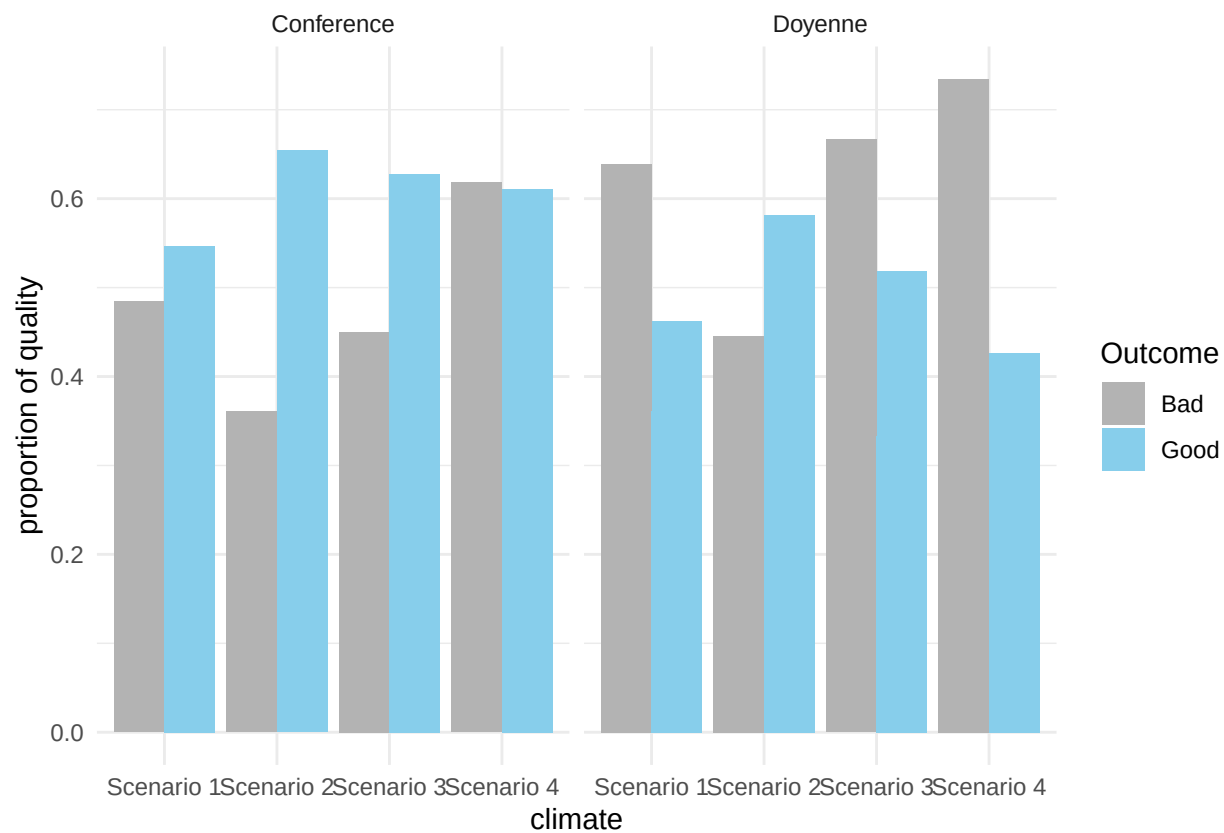
```
main.data2 %>%
  group_by(location, tree_id, species, climate) %>%
  summarise(prob_success = mean(quality_b), .groups = "drop") %>%
  ggplot(aes(location, prob_success, fill = species))+
  geom_boxplot()+
  stat_summary(fun = mean, geom = "point",
              position = position_dodge(width = 0.75))+
  facet_wrap(~climate)+
  labs(y = "probability of success") +
  theme_bw()
```



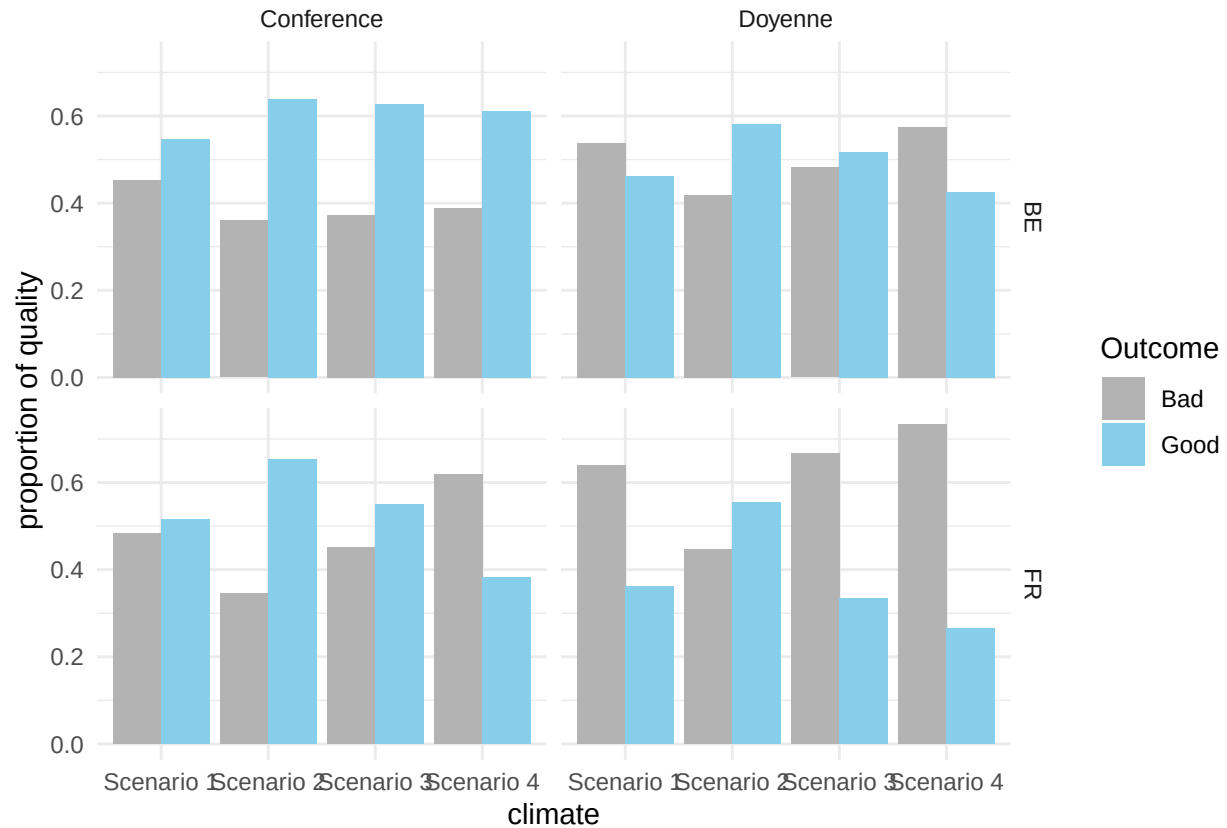
```
main.data2 %>%
  count(location, species, climate, quality_b) %>%
  group_by(location, species, climate) %>%
  mutate(percent = n / sum(n)) %>%
  ggplot(aes(climate, percent, fill = as.factor(quality_b))) +
  geom_bar(stat = "identity", position = "dodge") +
  facet_wrap(~location) +
  scale_fill_manual(values = c("gray70", "skyblue"), labels = c("Bad", "Good")) +
  labs(y = "proportion of quality", fill = "Outcome") +
  theme_minimal()
```



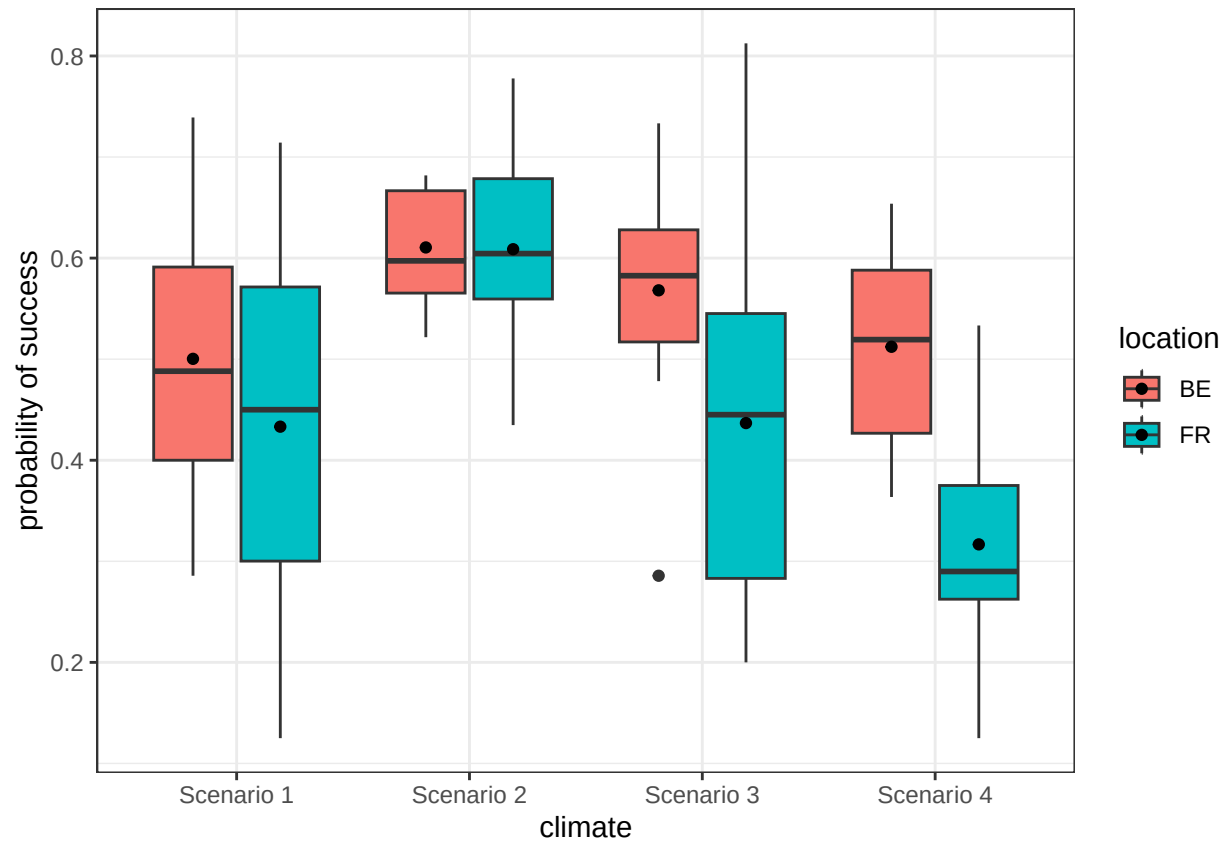
```
main.data2 %>%
  count(location, species, climate, quality_b) %>%
  group_by(location, species, climate) %>%
  mutate(percent = n / sum(n)) %>%
  ggplot(aes(climate, percent, fill = as.factor(quality_b))) +
  geom_bar(stat = "identity", position = "dodge") +
  facet_wrap(~species) +
  scale_fill_manual(values = c("gray70", "skyblue"), labels = c("Bad", "Good")) +
  labs(y = "proportion of quality", fill = "Outcome") +
  theme_minimal()
```



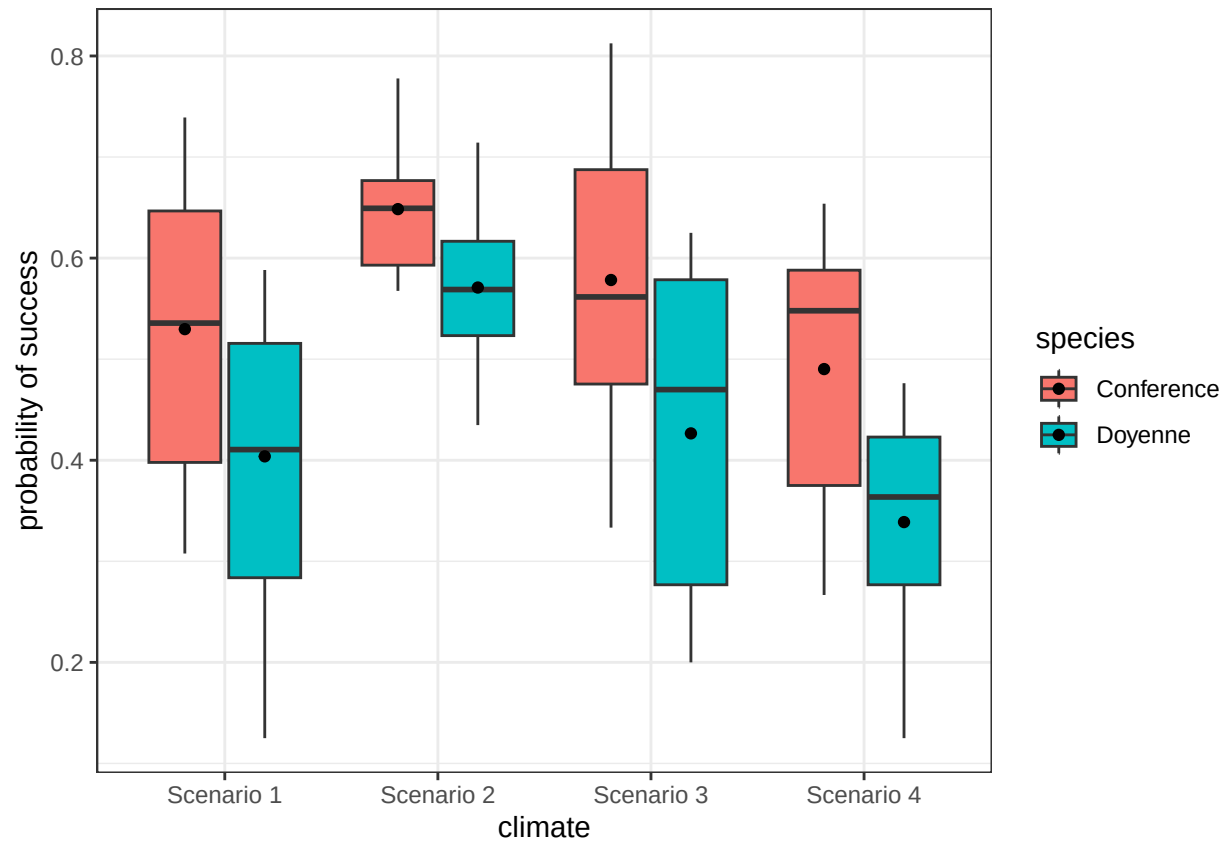
```
main.data2 %>%
  count(location, species, climate, quality_b) %>%
  group_by(location, species, climate) %>%
  mutate(percent = n / sum(n)) %>%
  ggplot(aes(climate, percent, fill = as.factor(quality_b))) +
  geom_bar(stat = "identity", position = "dodge") +
  facet_grid(location~species) +
  scale_fill_manual(values = c("gray70", "skyblue"), labels = c("Bad", "Good")) +
  labs(y = "proportion of quality", fill = "Outcome") +
  theme_minimal()
```



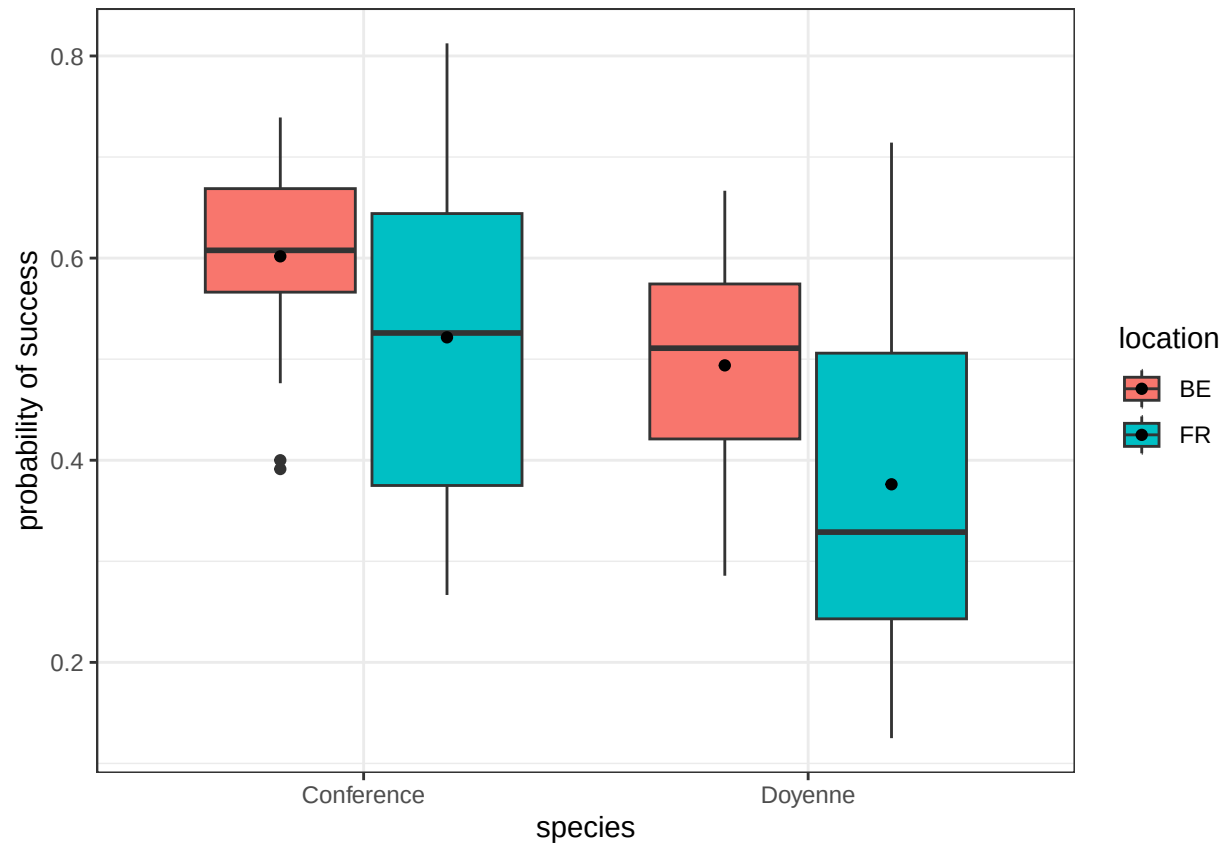
```
main.data2 %>%
  group_by(location, tree_id, species, climate) %>%
  summarise(prob_success = mean(quality_b), .groups = "drop") %>%
  ggplot(aes(climate, prob_success, fill = location)) +
  geom_boxplot() +
  stat_summary(fun = mean, geom = "point",
              position = position_dodge(width = 0.75)) +
  labs(y="probability of success") +
  theme_bw()
```



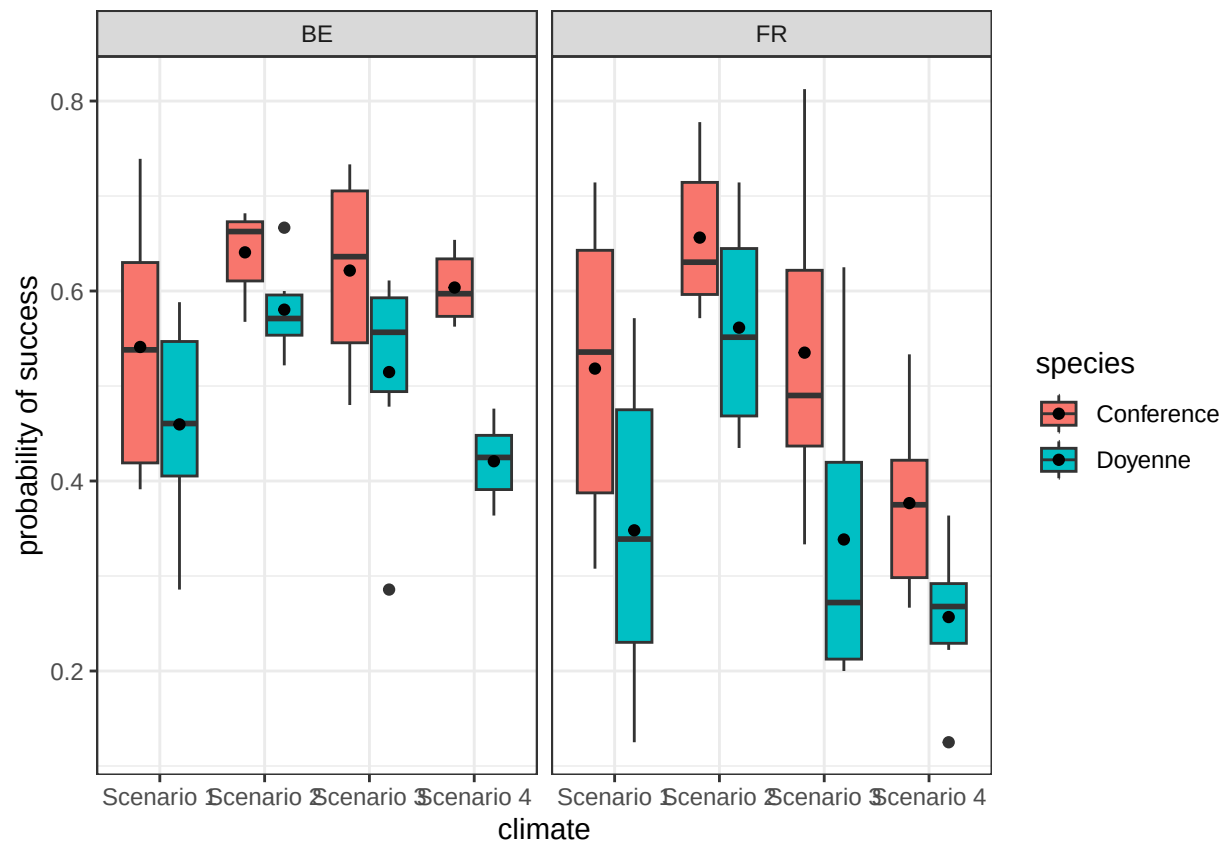
```
main.data2 %>%
  group_by(location, tree_id, species, climate) %>%
  summarise(prob_success = mean(quality_b), .groups = "drop") %>%
  ggplot(aes(climate, prob_success, fill = species)) +
  geom_boxplot() +
  stat_summary(fun = mean, geom = "point",
              position = position_dodge(width = 0.75)) +
  labs(y="probability of success") +
  theme_bw()
```



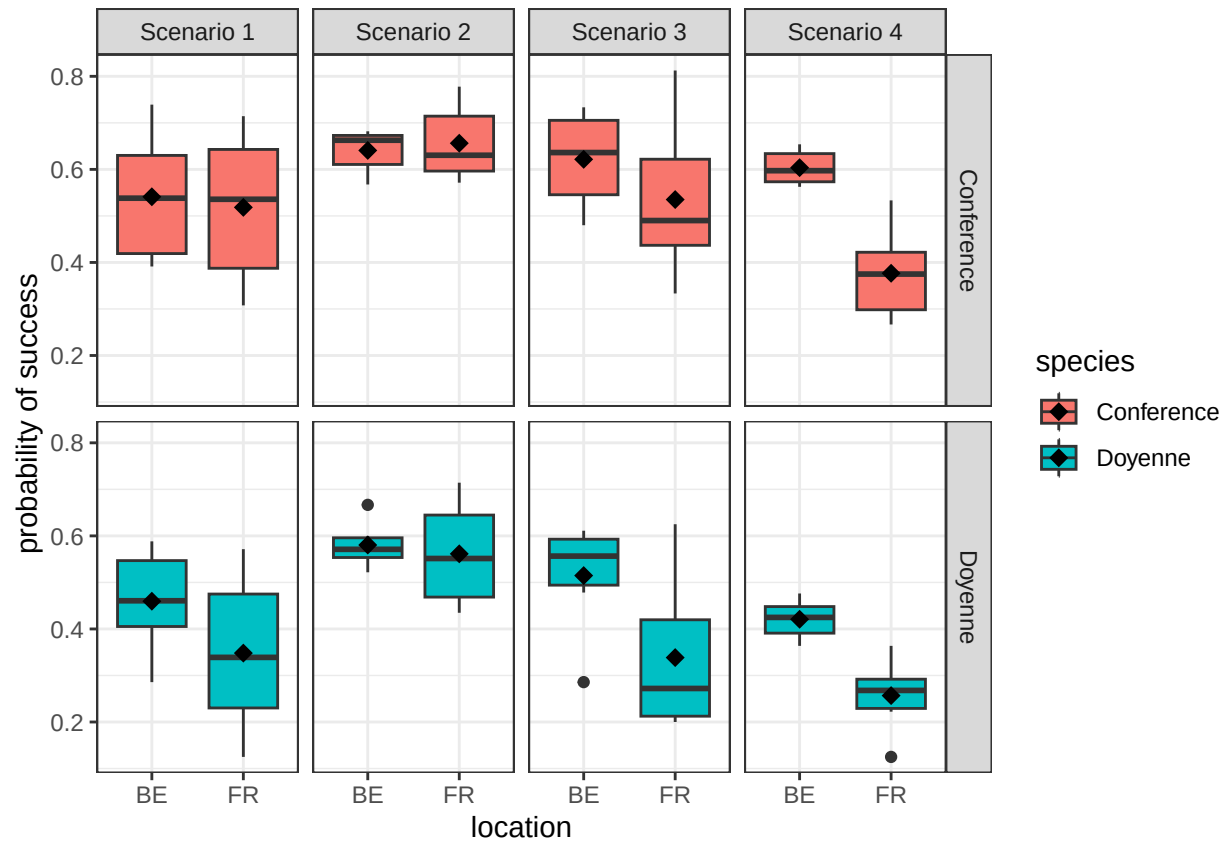
```
main.data2 %>%
  group_by(location, tree_id, species, climate) %>%
  summarise(prob_success = mean(quality_b), .groups = "drop") %>%
  ggplot(aes(species, prob_success, fill = location)) +
  geom_boxplot() +
  stat_summary(fun = mean, geom = "point",
              position = position_dodge(width = 0.75)) +
  labs(y="probability of success") +
  theme_bw()
```

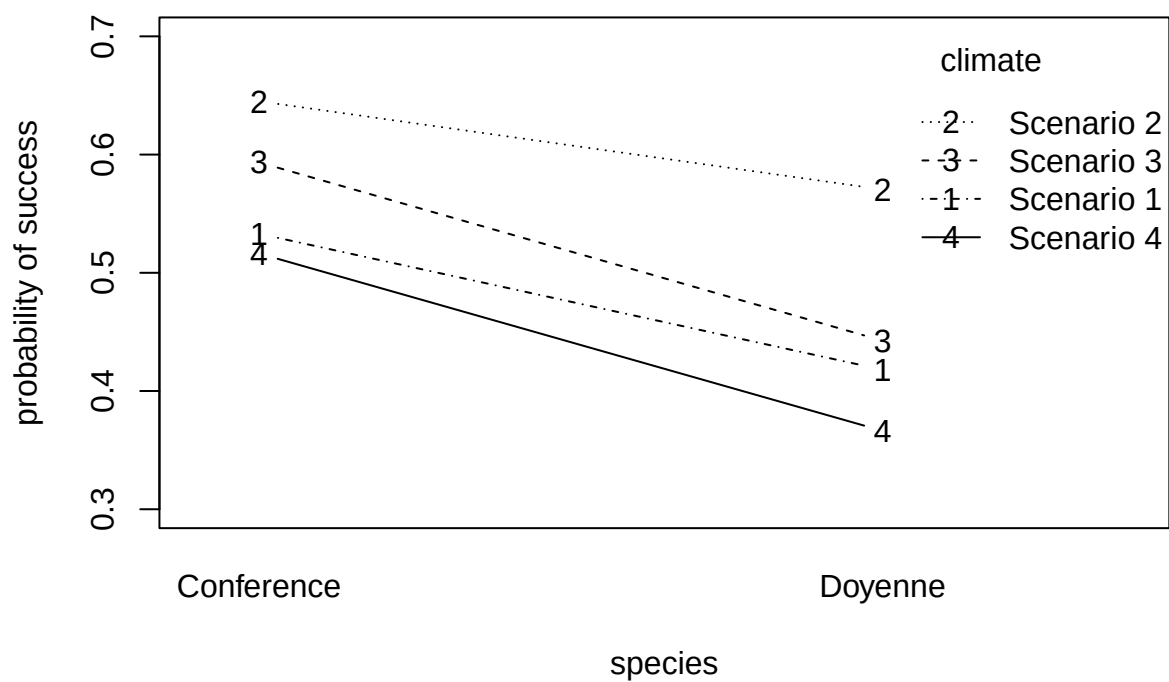
```
main.data2 %>%
  group_by(location, tree_id, species, climate) %>%
  summarise(prob_success = mean(quality_b), .groups = "drop") %>%
  ggplot(aes(climate, prob_success, fill = species)) +
  geom_boxplot() +
  stat_summary(fun = mean, geom = "point",
              position = position_dodge(width = 0.75)) +
  facet_wrap(~location) +
  labs(y="probability of success") +
  theme_bw()
```



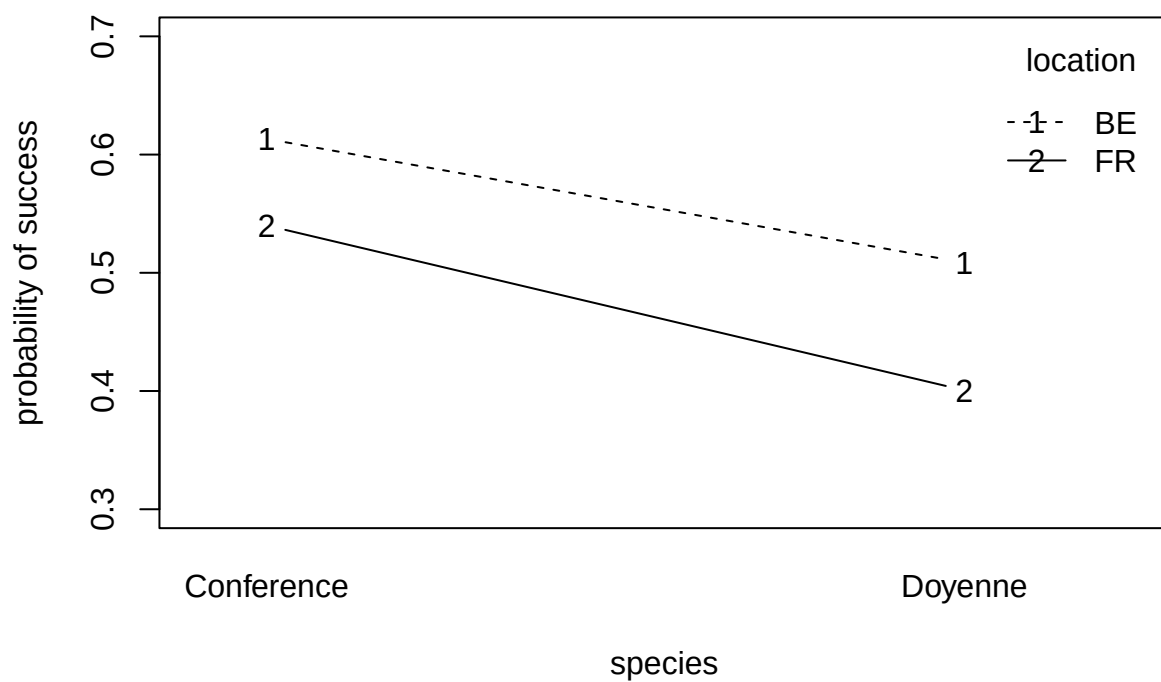
```
main.data2 %>%
  group_by(location, tree_id, species, climate) %>%
  summarise(prob_success = mean(quality_b), .groups = "drop") %>%
  ggplot(aes(x = location, y = prob_success, fill = species)) +
  geom_boxplot(position = position_dodge(width = 0.8)) +
  stat_summary(fun = mean, geom = "point", shape = 18, size = 3, color = "black",
    position = position_dodge(width = 0.8)) +
  facet_grid(species ~ climate) +
  labs(y = "probability of success") +
  theme_bw()
```



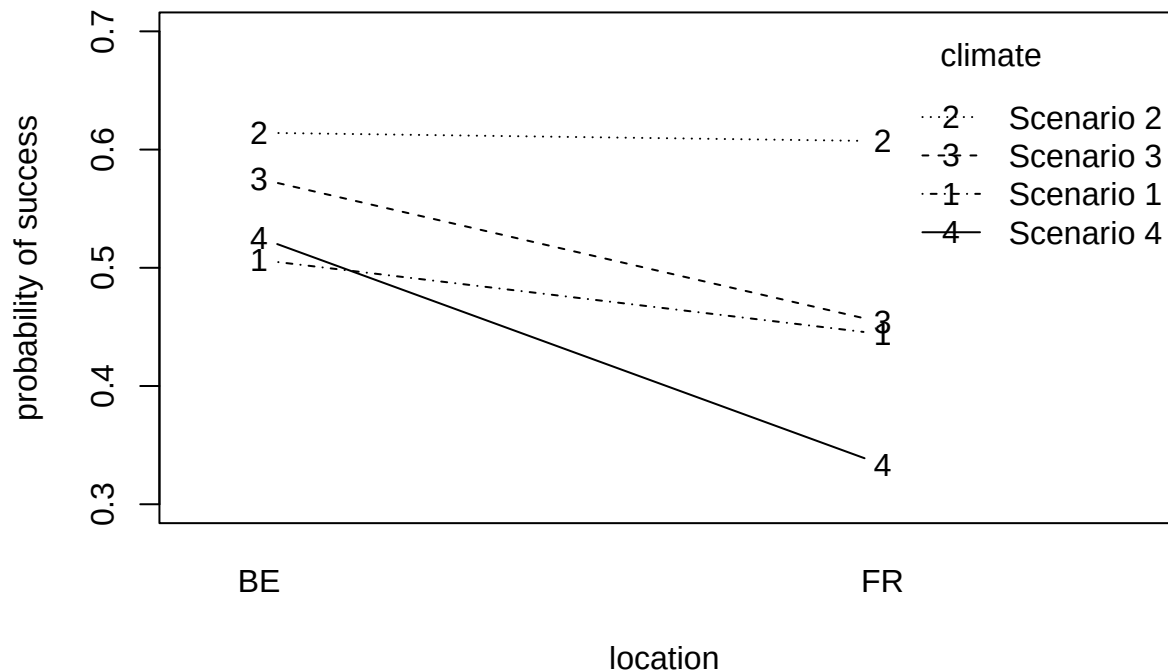
```
interaction.plot(x.factor = main.data2$species,
  trace.factor = main.data2$climate,
  response = main.data2$quality_b,
  trace.label = "climate",
  type = "b",
  xlab = "species",
  ylab = "probability of success",
  ylim = c(0.30, 0.7))
```



```
interaction.plot(x.factor = main.data2$species,
  trace.factor = main.data2$location,
  response = main.data2$quality_b,
  trace.label = "location",
  type = "b",
  xlab = "species",
  ylab = "probability of success",
  ylim = c(0.30, 0.7))
```



```
interaction.plot(x.factor = main.data2$location,  
                 trace.factor = main.data2$climate,  
                 response = main.data2$quality_b,  
                 trace.label = "climate",  
                 type = "b",  
                 xlab = "location",  
                 ylab = "probability of success",  
                 ylim = c(0.30, 0.7))
```



```
time.data <- read.csv("D:/UHasselt/2nd semester/PMHD/project/G12.size.data.csv",
                      sep = ",")
str(time.data)
```

```
## 'data.frame': 192 obs. of 25 variables:
## $ pear_id : int 1 2 3 4 5 6 7 8 9 10 ...
## $ tree_id : int 1 1 1 1 1 1 1 1 1 1 ...
## $ ecotr_id: int 1 1 1 1 1 1 1 1 1 1 ...
## $ species : chr "Conference" "Conference" "Conference" "Conference" ...
## $ climate : chr "Scenario 1" "Scenario 1" "Scenario 1" "Scenario 1" ...
## $ week_5 : num 3.69 4.09 3.51 3.56 4.02 ...
## $ week_6 : num 3.94 4.3 4.09 4.12 4.26 ...
## $ week_7 : num 4.59 4.66 4.47 4.61 4.7 ...
## $ week_8 : num 5 5.2 4.74 4.89 4.88 ...
## $ week_9 : num 5.52 5.84 5.01 5.12 5.37 ...
## $ week_10 : num 5.89 5.99 5.25 5.61 5.48 ...
## $ week_11 : num 5.98 6.18 5.49 5.83 5.68 ...
## $ week_12 : num 6.07 6.38 5.9 6.69 5.78 ...
## $ week_13 : num 6.2 6.61 6.31 6.72 6.58 ...
## $ week_14 : num 7.06 6.82 6.57 6.75 6.72 ...
## $ week_15 : num 7.36 7.76 6.83 6.78 6.87 ...
## $ week_16 : num 7.57 8.03 7.09 6.81 6.89 ...
## $ week_17 : num 7.74 8.29 7.66 6.92 7.66 ...
## $ week_18 : num 7.9 8.56 7.69 7.24 7.99 ...
## $ week_19 : num 8.13 8.79 7.72 7.56 8.32 ...
## $ week_20 : num 8.5 8.84 7.76 8.03 8.89 ...
```

```
## $ week_21 : num 8.75 8.89 7.92 8.49 8.96 ...
## $ week_22 : num 9.29 9.49 8.38 8.87 9.48 ...
## $ week_23 : num NA 9.82 8.6 8.87 9.92 ...
## $ week_24 : num NA 10.19 8.96 8.87 10.35 ...
```

```
sapply(time.data, function(x) sum(is.na(x)))
```

```
## pear_id tree_id ecotr_id species climate week_5 week_6 week_7
##      0      0      0      0      0      0      0      0
## week_8 week_9 week_10 week_11 week_12 week_13 week_14 week_15
##      2      5      7      8      9     10     10     13
## week_16 week_17 week_18 week_19 week_20 week_21 week_22 week_23
##     13     13     14     14     15     15     17     19
## week_24
##      19
```

```
time.data[!complete.cases(time.data), ]
```

```
##      pear_id tree_id ecotr_id species climate week_5 week_6 week_7
## 1          1      1      1 Conference Scenario 1 3.693860 3.942590 4.587824
## 8          8      1      1 Conference Scenario 1 4.116225 4.445900 4.775575
## 32         32      2      1 Doyenne Scenario 1 3.142866 3.273142 3.847718
## 36         36      2      1 Doyenne Scenario 1 3.656709 4.269728 4.376240
## 39         39      2      1 Doyenne Scenario 1 2.932664 3.202827 3.858217
## 48         48      2      1 Doyenne Scenario 1 3.004247 3.671863 3.826864
## 73         73      4      2 Doyenne Scenario 2 4.027888 4.195575 4.363262
## 75         75      4      2 Doyenne Scenario 2 3.599844 3.747326 3.894808
## 79         79      4      2 Doyenne Scenario 2 3.413932 3.462084 4.376531
## 81         81      4      2 Doyenne Scenario 2 3.284493 3.811102 4.280054
## 92         92      4      2 Doyenne Scenario 2 3.825882 3.873091 3.920301
## 115        115      5      3 Conference Scenario 3 3.406182 3.989412 4.347570
## 130        130      6      3 Doyenne Scenario 3 3.704906 3.780380 3.855855
## 138        138      6      3 Doyenne Scenario 3 3.311664 3.848212 3.851519
## 140        140      6      3 Doyenne Scenario 3 3.029741 3.468400 3.852316
## 144        144      7      4 Conference Scenario 4 3.698578 3.776706 4.198246
## 148        148      7      4 Conference Scenario 4 3.428523 3.876310 4.043211
## 155        155      7      4 Conference Scenario 4 4.115689 4.406696 4.697703
## 161        161      7      4 Conference Scenario 4 4.259865 4.390626 4.441801
##      week_8 week_9 week_10 week_11 week_12 week_13 week_14 week_15
## 1 4.995353 5.515952 5.889033 5.979142 6.069251 6.198874 7.061182 7.357664
## 8 5.017853      NA      NA      NA      NA      NA      NA      NA
## 32 4.060115 4.443438 4.480119 4.516799 4.899601 4.949316 5.761092      NA
## 36 4.482752 4.589263 4.724220 4.859177 4.994134 5.129091 5.594412      NA
## 39 4.121770 4.385323 4.487525 4.527822 4.696921 4.866020 5.266573 5.493700
## 48 3.981865 4.136866 4.488428 4.839990 4.863237 4.886484 4.909731      NA
## 73 4.530949 4.698635 5.214744 5.730853 5.770216 5.809580 5.848944 5.965977
## 75 4.087929 4.281049 4.643193 4.831493 4.923023      NA      NA      NA
## 79 4.415890      NA      NA      NA      NA      NA      NA      NA
## 81 4.674167 5.068280      NA      NA      NA      NA      NA      NA
## 92 3.967510 5.197322 5.397216 5.597111 5.797005 5.832262 5.867519 6.163430
## 115 4.862282      NA      NA      NA      NA      NA      NA      NA
## 130 4.266879 4.391457 4.516035 4.824134 5.132234 5.653480 5.771910 5.834205
## 138 3.986296 4.185514 4.384731 4.579258 4.983305 5.347759 5.692837 6.037914
## 140 3.926675 4.363219 4.586496 5.004238      NA      NA      NA      NA
## 144      NA      NA      NA      NA      NA      NA      NA      NA
```

```
## 148 4.519133 4.847104 4.981541      NA      NA      NA      NA      NA
## 155      NA      NA      NA      NA      NA      NA      NA      NA      NA
## 161 4.573859 5.108958      NA      NA      NA      NA      NA      NA      NA
##      week_16 week_17 week_18 week_19 week_20 week_21 week_22 week_23
## 1    7.570166 7.736439 7.902711 8.128065 8.499614 8.747300 9.293133      NA
## 8      NA      NA      NA      NA      NA      NA      NA      NA      NA
## 32      NA      NA      NA      NA      NA      NA      NA      NA      NA
## 36      NA      NA      NA      NA      NA      NA      NA      NA      NA
## 39    5.668764 6.361651 6.599155 7.164853 7.175148 7.185443      NA      NA
## 48      NA      NA      NA      NA      NA      NA      NA      NA      NA
## 73    6.083010 6.951444 7.247997 7.544550      NA      NA      NA      NA
## 75      NA      NA      NA      NA      NA      NA      NA      NA      NA
## 79      NA      NA      NA      NA      NA      NA      NA      NA      NA
## 81      NA      NA      NA      NA      NA      NA      NA      NA      NA
## 92    6.828448 6.876572 6.924696 7.291133 7.580955 8.245821      NA      NA
## 115     NA      NA      NA      NA      NA      NA      NA      NA      NA
## 130    5.847786 6.035299      NA      NA      NA      NA      NA      NA      NA
## 138    6.169943 6.320564 6.406123 6.520879 6.635635 6.864247 7.303086      NA
## 140     NA      NA      NA      NA      NA      NA      NA      NA      NA
## 144     NA      NA      NA      NA      NA      NA      NA      NA      NA
## 148     NA      NA      NA      NA      NA      NA      NA      NA      NA
## 155     NA      NA      NA      NA      NA      NA      NA      NA      NA
## 161     NA      NA      NA      NA      NA      NA      NA      NA      NA
##      week_24
## 1      NA
## 8      NA
## 32     NA
## 36     NA
## 39     NA
## 48     NA
## 73     NA
## 75     NA
## 79     NA
## 81     NA
## 92     NA
## 115    NA
## 130    NA
## 138    NA
## 140    NA
## 144    NA
## 148    NA
## 155    NA
## 161    NA
```

```
library(naniar)
```

```
## Warning: package 'naniar' was built under R version 4.5.3
```

```
miss_var_summary(time.data)
```

```
## # A tibble: 25 x 3
##   variable n_miss pct_miss
##   <chr>     <int>   <num>
## 1 week_23      19     9.90
## 2 week_24      19     9.90
```



```
## 3 week_22      17      8.85
## 4 week_20      15      7.81
## 5 week_21      15      7.81
## 6 week_18      14      7.29
## 7 week_19      14      7.29
## 8 week_15      13      6.77
## 9 week_16      13      6.77
## 10 week_17     13      6.77
## # i 15 more rows
```

there are some drop out data, not missing. Some observations were obtained but dropout after a certain points.

```
time.data2 <- reshape(time.data, direction = "long",
  varying = c("week_5", "week_6", "week_7", "week_8", "week_9",
    "week_10", "week_11", "week_12", "week_13", "week_14",
    "week_15", "week_16", "week_17", "week_18", "week_19",
    "week_20", "week_21", "week_22", "week_23", "week_24"),
  idvar = c("pear_id", "tree_id", "ecotr_id", "species", "climate"),
  timevar = "time", times = 5:24, v.names = "quality")
str(time.data2)
```

```
## 'data.frame': 3840 obs. of 7 variables:
## $ pear_id : int 1 2 3 4 5 6 7 8 9 10 ...
## $ tree_id : int 1 1 1 1 1 1 1 1 1 1 ...
## $ ecotr_id: int 1 1 1 1 1 1 1 1 1 1 ...
## $ species : chr "Conference" "Conference" "Conference" "Conference" ...
## $ climate : chr "Scenario 1" "Scenario 1" "Scenario 1" "Scenario 1" ...
## $ time : int 5 5 5 5 5 5 5 5 5 5 ...
## $ quality : num 3.69 4.09 3.51 3.56 4.02 ...
## - attr(*, "reshapeLong")=List of 4
## ..$ varying:List of 1
## .. ..$ quality: chr [1:20] "week_5" "week_6" "week_7" "week_8" ...
## .. ..- attr(*, "v.names")= chr "quality"
## .. ..- attr(*, "times")= int [1:20] 5 6 7 8 9 10 11 12 13 14 ...
## ..$ v.names: chr "quality"
## ..$ idvar : chr [1:5] "pear_id" "tree_id" "ecotr_id" "species" ...
## ..$ timevar: chr "time"
```

```
# write.csv(time.data2, "D:/UHasselt/2nd semester/PMHD/project/longitudinal_data.csv",
#           row.names = FALSE)
```

```
time.data2 %>%
  group_by(species) %>%
  summarise(Total_NA = sum(is.na(quality)))
```

```
## # A tibble: 2 x 2
##   species    Total_NA
##   <chr>      <int>
## 1 Conference      97
## 2 Doyenne       106
```

```
dropout_patterns <- time.data2 %>%
  filter(!is.na(quality)) %>%
  group_by(pear_id, species, climate) %>%
  summarise(last_obs_week = max(time), .groups = "drop")
```

```

baseline <- dropout_patterns %>%
  count(species, climate, name = "N")

risk_data <- dropout_patterns %>%
  crossing(Week = 5:24) %>%
  group_by(species, climate, Week) %>%
  summarise(n = sum(last_obs_week >= Week), .groups = "drop") %>%
  left_join(baseline, by = c("species", "climate")) %>%
  mutate(prop = n / N,
         percent = round(100 * prop))

table_14_1 <- risk_data %>%
  mutate(value = paste0(n, " (", percent, "%")) %>%
  select(Week, species, value) %>%
  pivot_wider(names_from = species, values_from = value)

```

```

## Warning: Values from `value` are not uniquely identified; output will contain list-cols.
## * Use `values_fn = list` to suppress this warning.
## * Use `values_fn = {summary_fun}` to summarise duplicates.
## * Use the following dplyr code to identify duplicates.
## {data} |>
## dplyr::summarise(n = dplyr::n(), .by = c(Week, species)) |>
## dplyr::filter(n > 1L)

```

```

risk_data %>%
  group_by(species, Week) %>%
  summarise(n = sum(n),
            N = sum(N),
            percent = round(100 * n / N),
            .groups = "drop") %>%
  mutate(value = paste0(n, " (", percent, "%")) %>%
  select(Week, species, value) %>%
  pivot_wider(names_from = species, values_from = value)

```

```

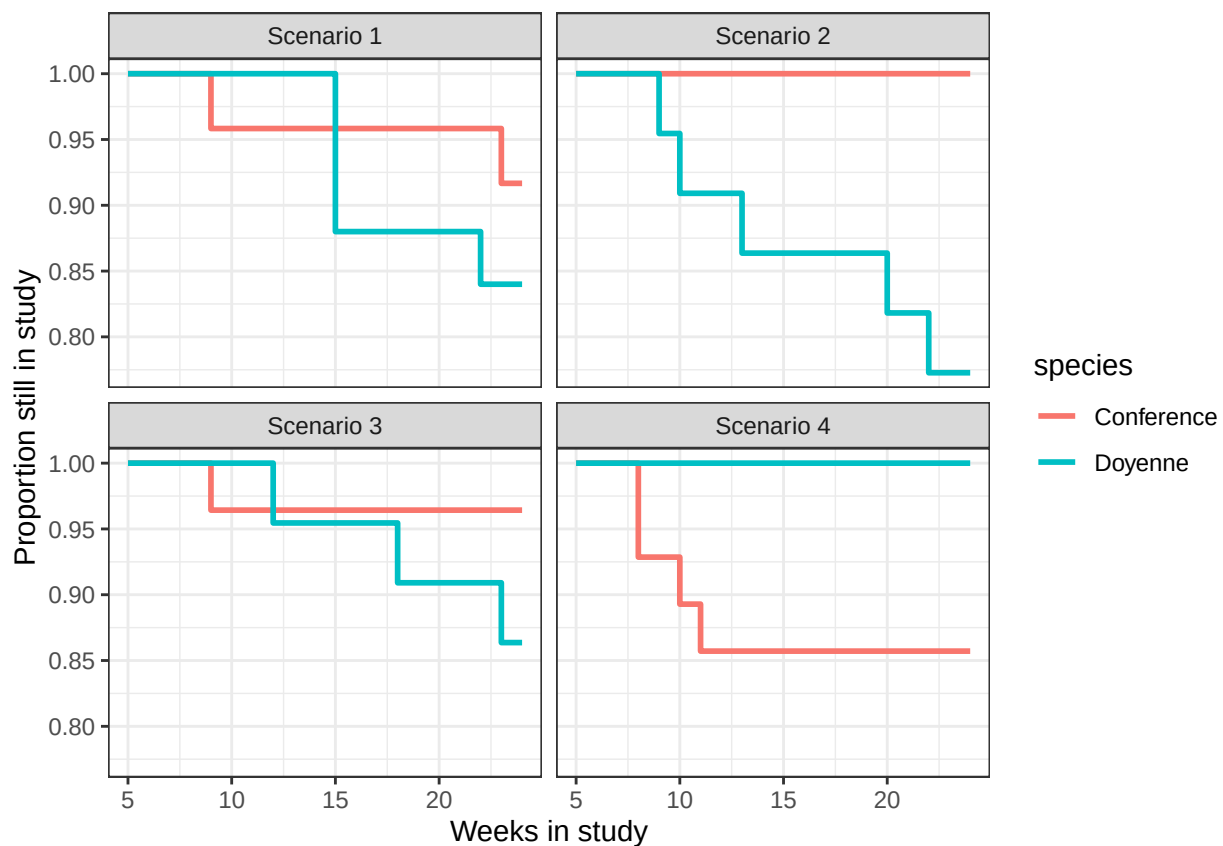
## # A tibble: 20 x 3
##   Week Conference Doyenne
##   <int> <chr>      <chr>
## 1     5 101 (100)  91 (100)
## 2     6 101 (100)  91 (100)
## 3     7 101 (100)  91 (100)
## 4     8  99 (98)   91 (100)
## 5     9  97 (96)   90 (99)
## 6    10  96 (95)   89 (98)
## 7    11  95 (94)   89 (98)
## 8    12  95 (94)   88 (97)
## 9    13  95 (94)   87 (96)
## 10   14  95 (94)   87 (96)
## 11   15  95 (94)   84 (92)
## 12   16  95 (94)   84 (92)
## 13   17  95 (94)   84 (92)
## 14   18  95 (94)   83 (91)
## 15   19  95 (94)   83 (91)
## 16   20  95 (94)   82 (90)
## 17   21  95 (94)   82 (90)

```

```
## 18    22 95 (94)    80 (88)
## 19    23 94 (93)    79 (87)
## 20    24 94 (93)    79 (87)
```

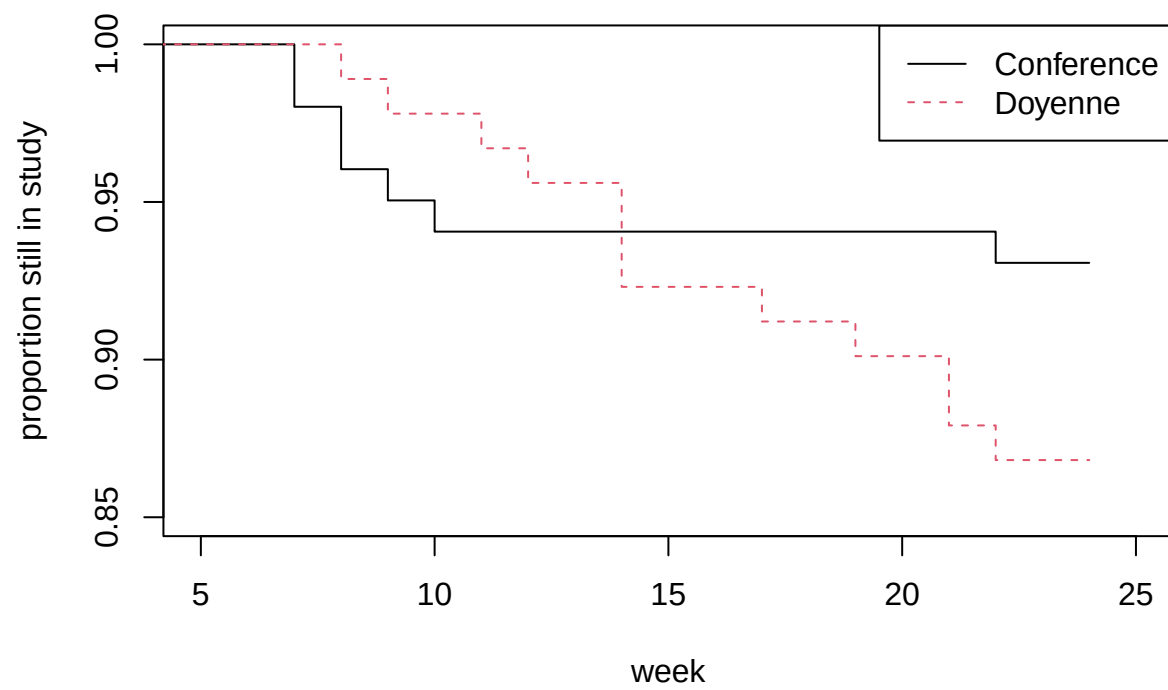
```
ggplot(risk_data, aes(x = Week, y = prop, color = species)) +
  geom_step(size = 1) +
  facet_wrap(~ climate) +
  labs(x = "Weeks in study", y = "Proportion still in study") +
  theme_bw()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



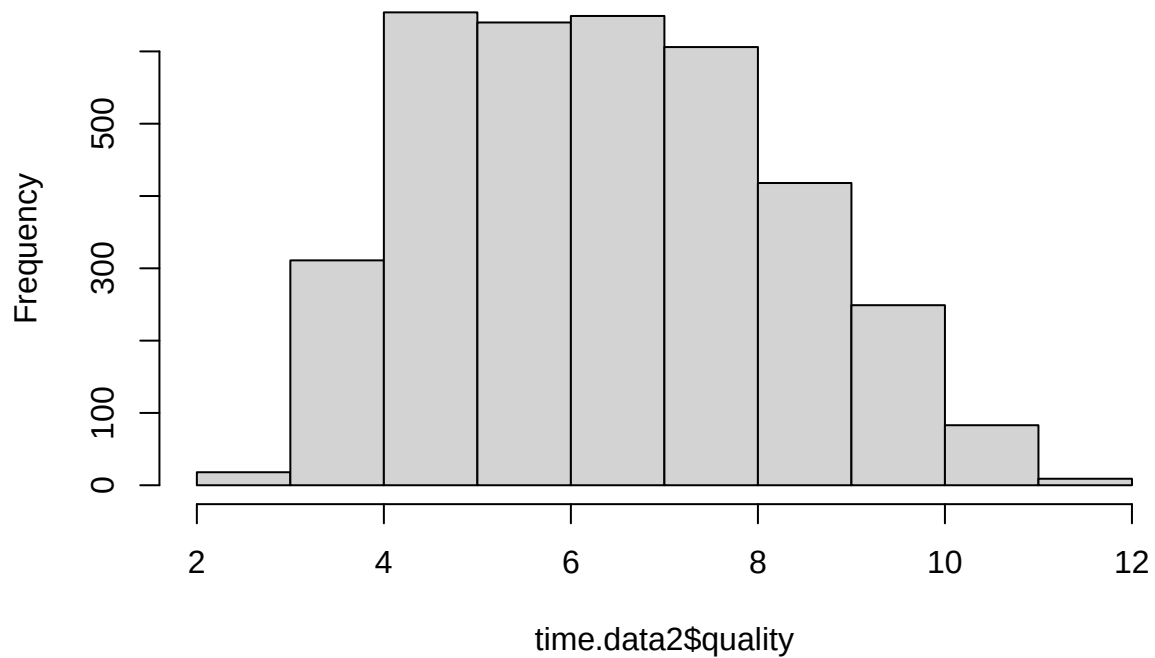
```
library(survival)
surv_obj <- Surv(
  time = dropout_patterns$last_obs_week,
  event = dropout_patterns$last_obs_week < 24)
fit <- survfit(surv_obj ~ species, data = dropout_patterns)

plot(fit, col = 1:2, lty = 1:2,
     xlim = c(5,25), ylim = c(0.85, 1.0),
     xlab = "week", ylab = "proportion still in study")
legend("topright", legend = unique(dropout_patterns$species),
     col = 1:2, lty = 1:2)
```



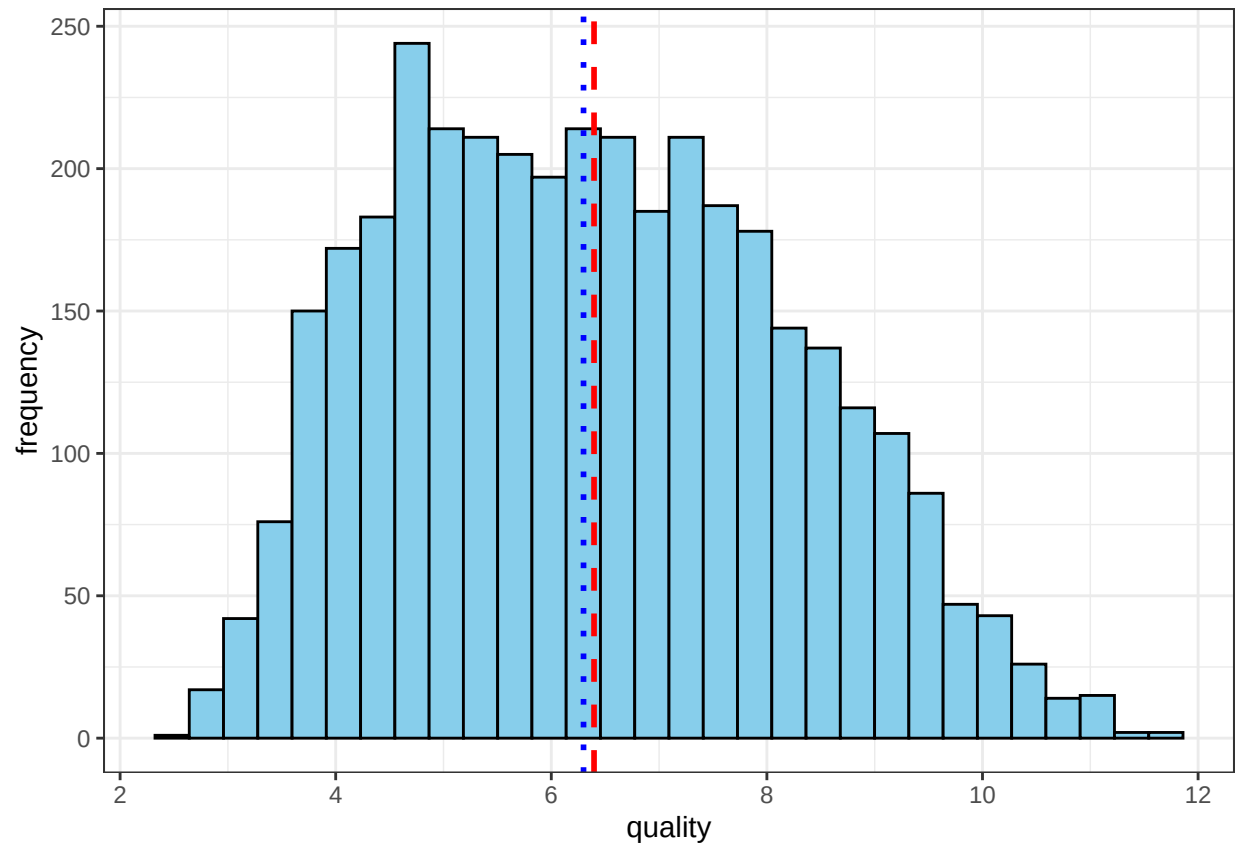
```
hist(time.data2$quality)
```

Histogram of time.data2\$quality



```
ggplot(time.data2) +  
  geom_histogram(aes(quality), bins = 30, fill = "skyblue", colour = "black") +  
  geom_vline(aes(xintercept = mean(time.data2$quality, na.rm=TRUE)), color = "red",  
             linetype = "dashed", size = 1) +  
  geom_vline(aes(xintercept = median(time.data2$quality, na.rm=TRUE)), color = "blue",  
             linetype = "dotted", size = 1) +  
  labs(y="frequency") +  
  scale_x_continuous(breaks = sort(c(pretty(time.data2$quality)))) +  
  theme_bw()
```

```
## Warning: Removed 203 rows containing non-finite outside the scale range  
## (`stat_bin()`).
```

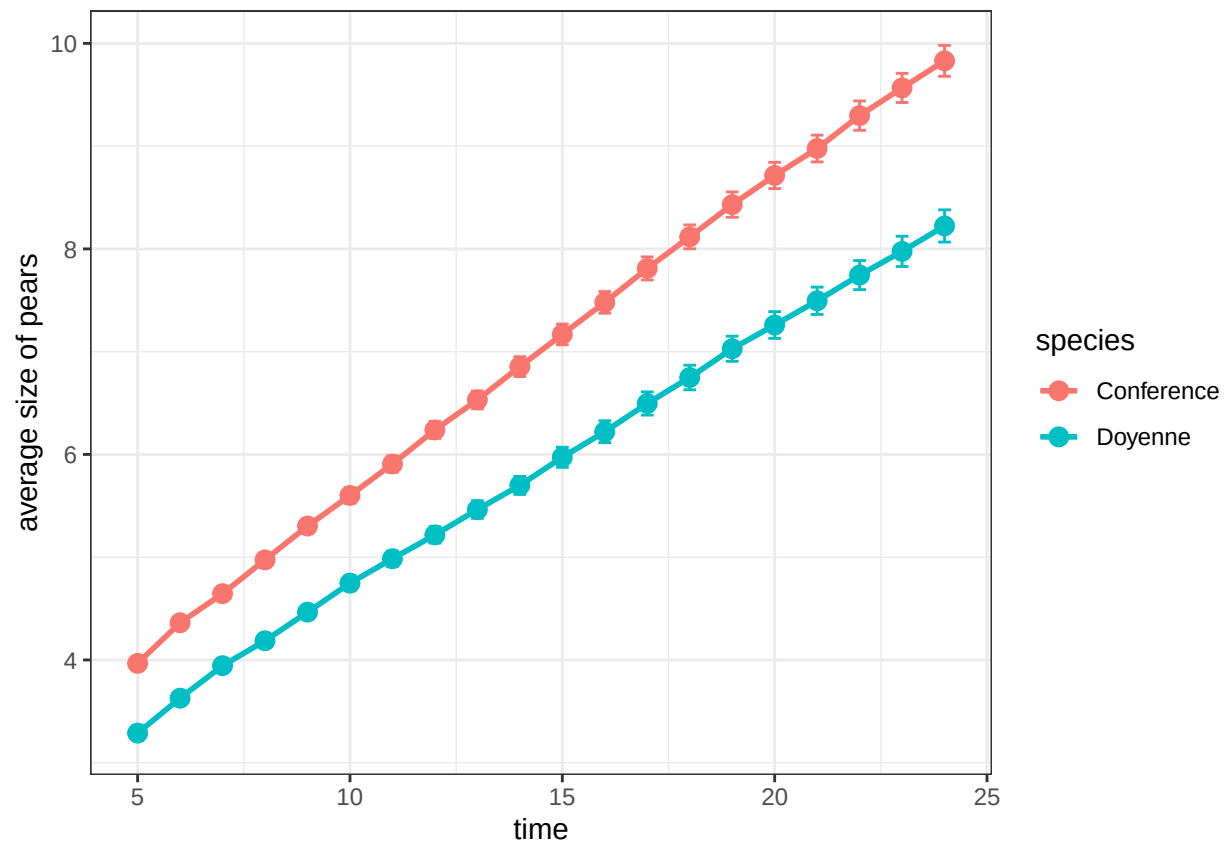


```
ggplot(time.data2, aes(x = time, y = quality, color = species, fill = species)) +
  stat_summary(fun = mean, geom = "line", size = 1) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.3) +
  labs(y="average size of pears") +
  theme_bw()
```

```
## Warning: Removed 203 rows containing non-finite outside the scale range
## (`stat_summary()`).
```

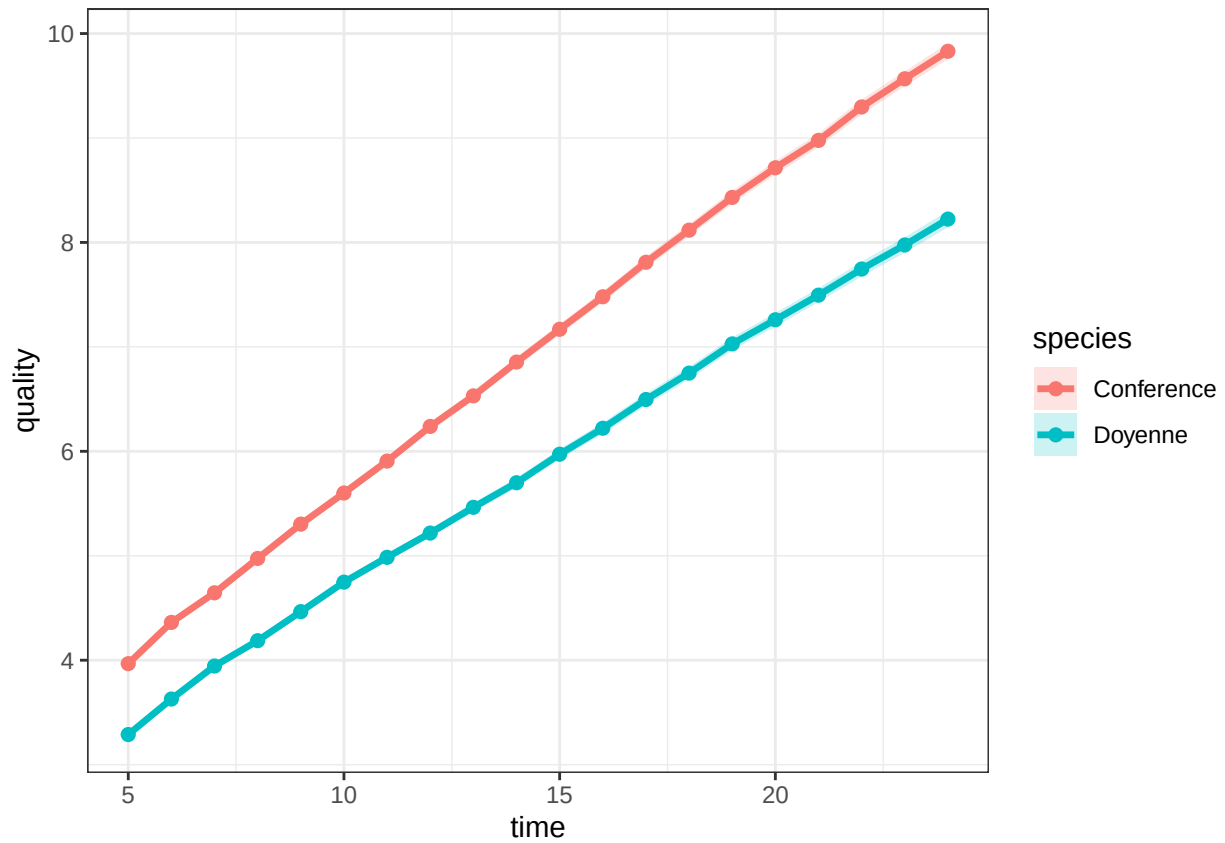
```
## Warning: Removed 203 rows containing non-finite outside the scale range
## (`stat_summary()`).
```

```
## Removed 203 rows containing non-finite outside the scale range
## (`stat_summary()`).
```



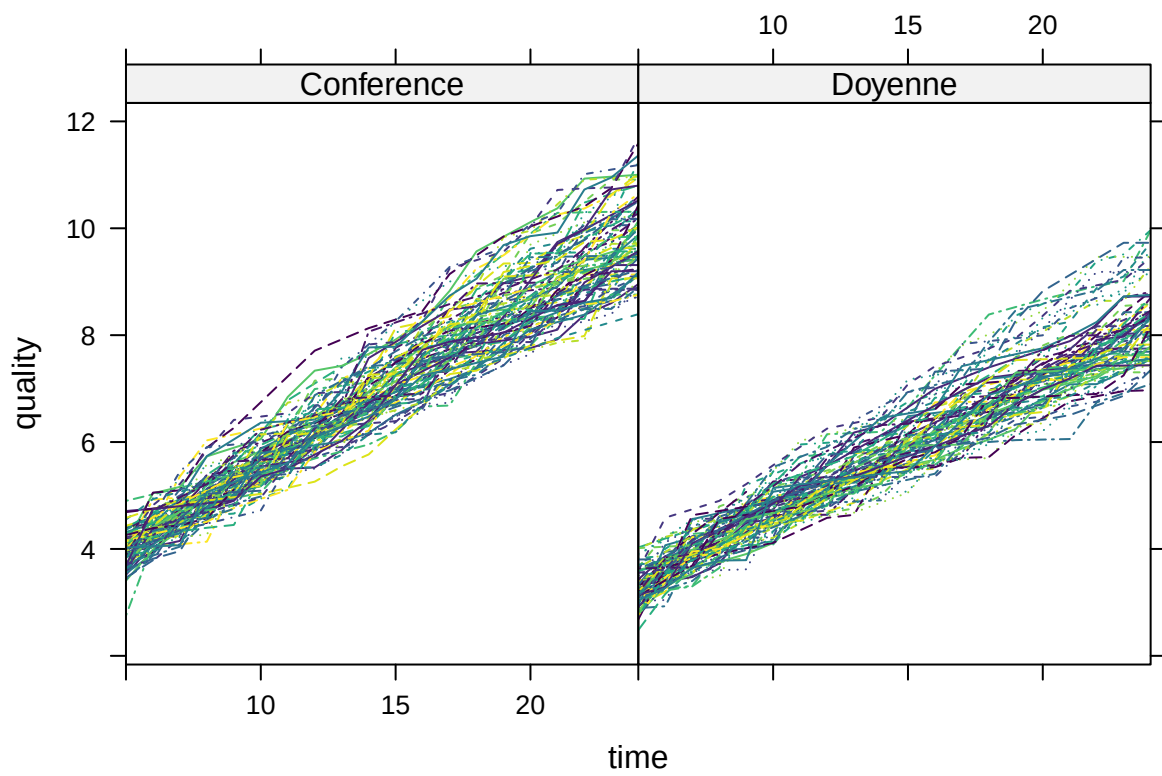
```
ggplot(time.data2, aes(time,quality,colour = species))+
  stat_summary(fun = mean, geom = "line", size = 1.2) +
  stat_summary(fun = mean, geom = "point", size = 2) +
  stat_summary(fun.data = mean_se, geom = "ribbon", aes(fill = species),
    alpha = 0.2, colour = NA) +
  theme_bw()
```

```
## Warning: Removed 203 rows containing non-finite outside the scale range
## (`stat_summary()`).
## Removed 203 rows containing non-finite outside the scale range
## (`stat_summary()`).
## Removed 203 rows containing non-finite outside the scale range
## (`stat_summary()`).
```

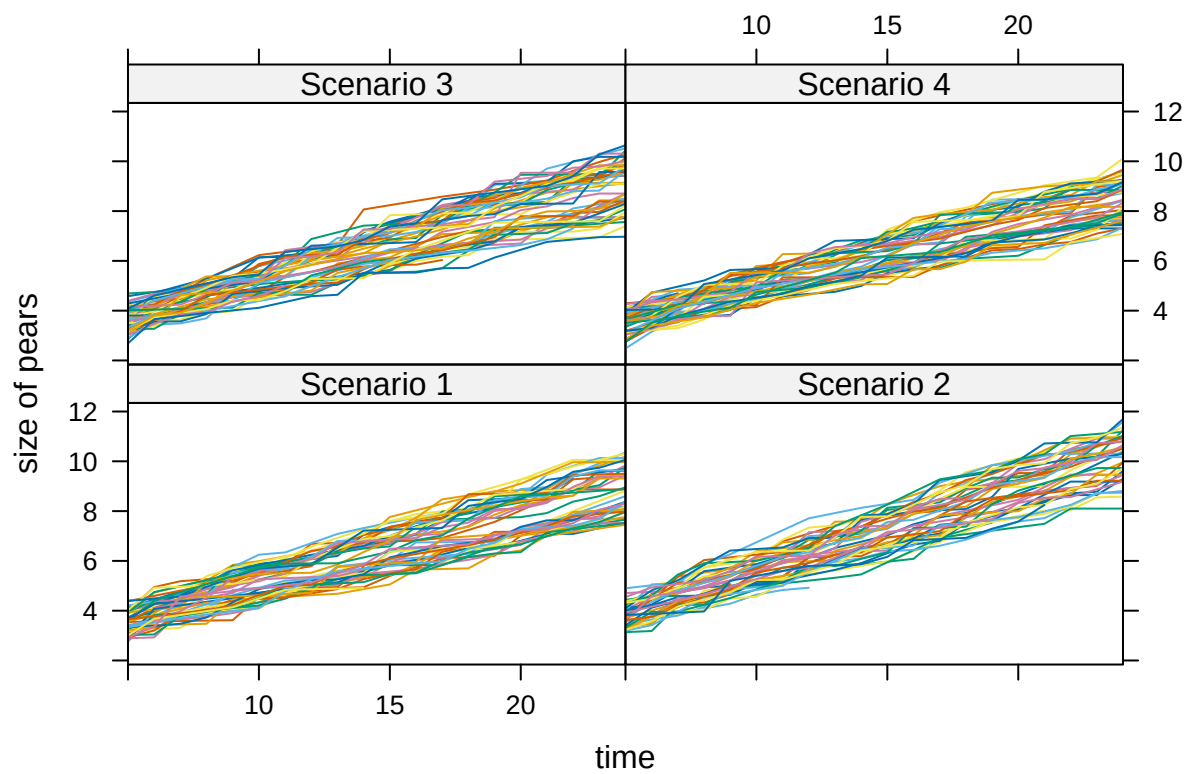


```
my_cols <- c("salmon", "steelblue", "darkgrey", "darkgreen")
my_colors<-viridis(20)
my_lty <-c(5:24)
my_pch <-c(5:24)
```

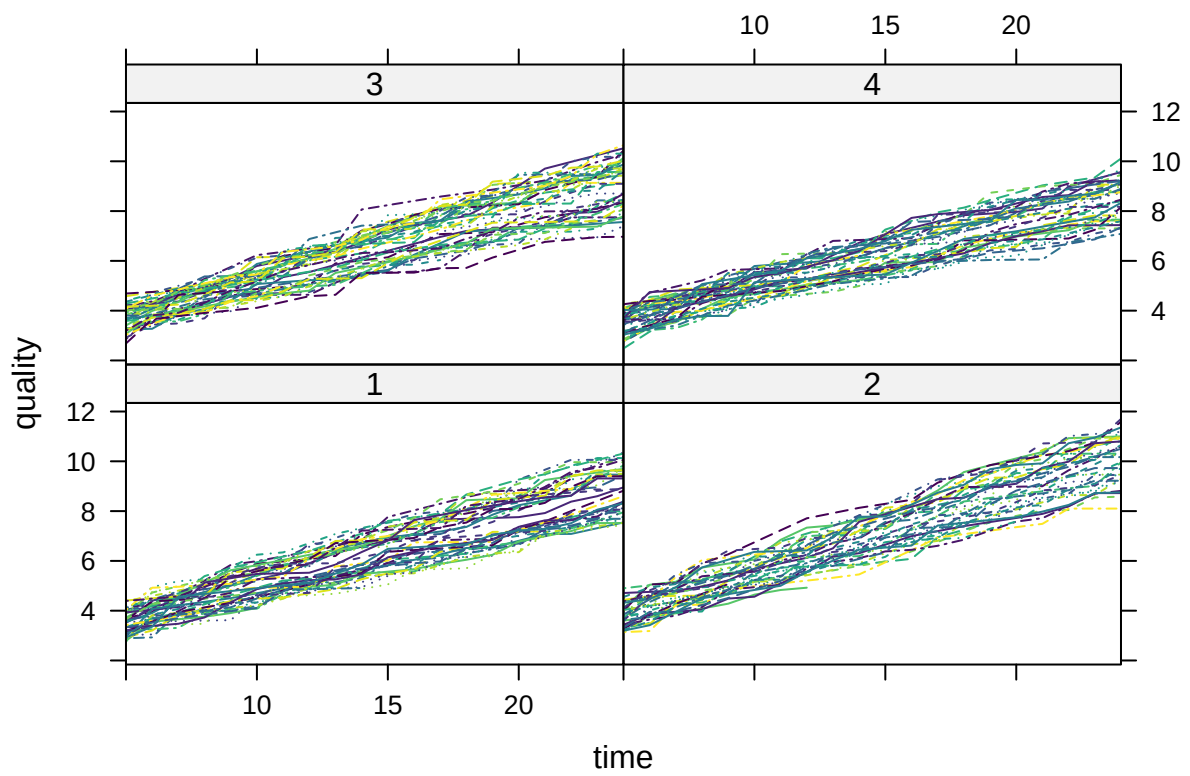
```
xyplot(quality ~ time | species,
       groups = pear_id,
       type = "l",
       col = my_colors,
       lty = my_lty,
       xlim = c(5, 24),
       data = time.data2)
```

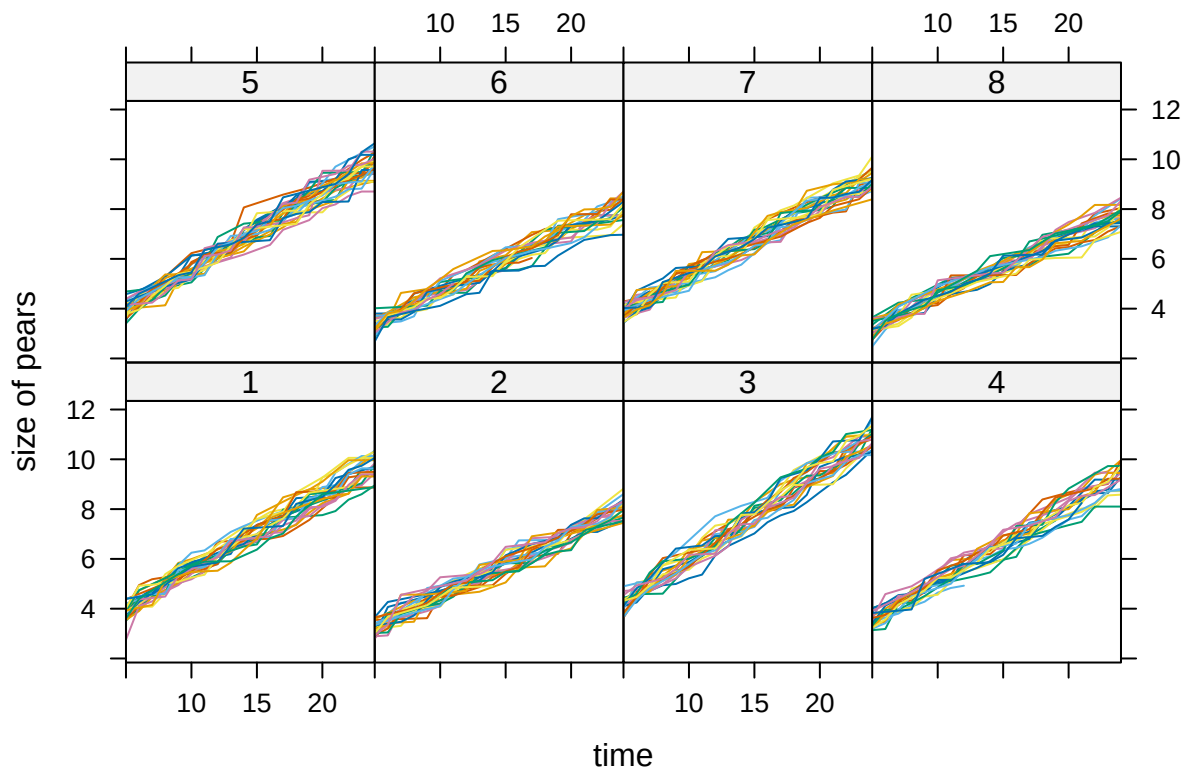
```
xyplot(quality ~ time | climate,
       groups = pear_id,
       type = "l",
       xlim = c(5, 24),
       ylab = "size of pears",
       data = time.data2)
```



```
xyplot(quality ~ time | as.factor(ecotr_id),
  groups = pear_id,
  type = "l",
  col = my_colors,
  lty = my_lty,
  xlim = c(5, 24),
  data = time.data2)
```



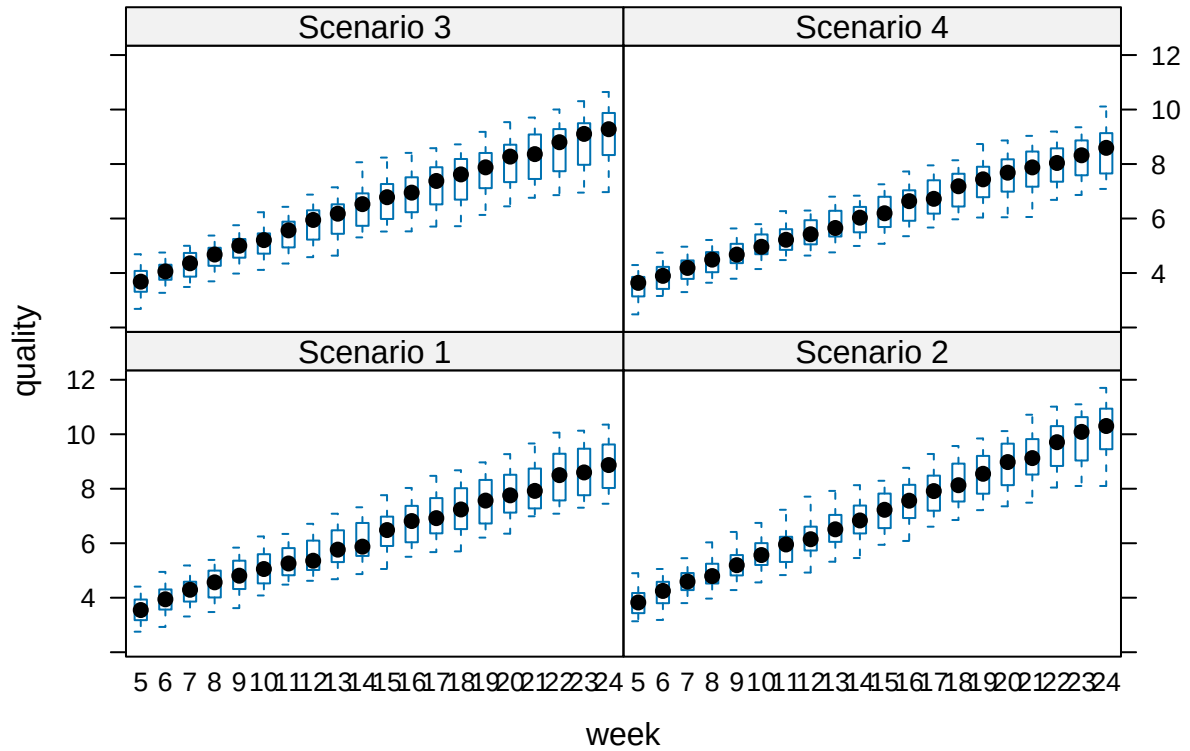
```
xyplot(quality ~ time | as.factor(tree_id),
       groups = pear_id,
       type = "l",
       xlim = c(5, 24),
       ylab = "size of pears",
       data = time.data2)
```



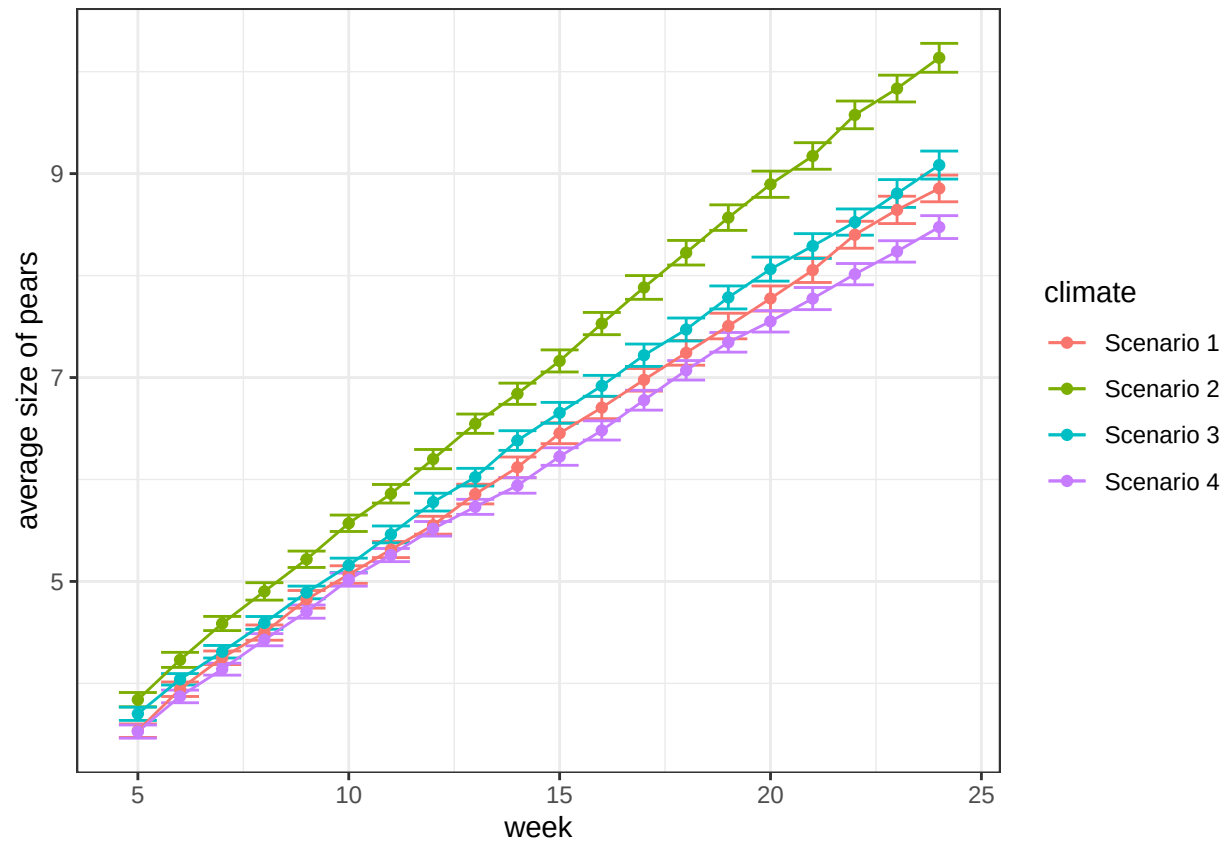
```
table(time.data2$ecotr_id,time.data2$tree_id)
```

```
##
##      1  2  3  4  5  6  7  8
## 1 480 500  0  0  0  0  0  0
## 2  0  0 420 440  0  0  0  0
## 3  0  0  0  0 560 440  0  0
## 4  0  0  0  0  0  0 560 440
```

```
bwplot(quality ~ as.factor(time) | climate,
       data = time.data2,
       xlab = "week",
       ylab = "quality",
       )
```

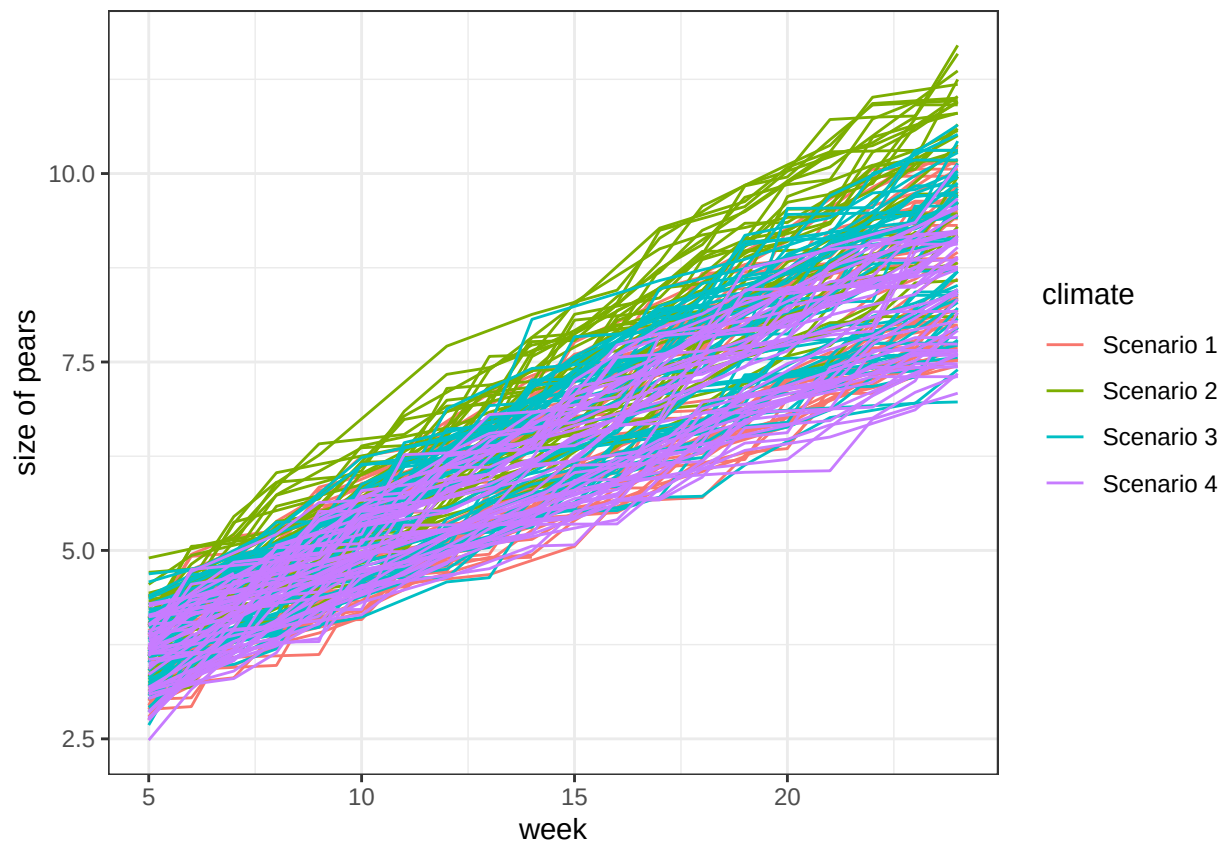


```
time.data2 %>%
  group_by(time, ecotr_id, climate) %>%
  summarise(mean_quality = mean(quality, na.rm = TRUE),
            sd_quality = sd(quality, na.rm = TRUE),
            n_obs = n(),
            se_quality = sd_quality / sqrt(n_obs),
            .groups = "drop") %>%
  ggplot(aes(x = time, y = mean_quality, colour = climate)) +
  geom_errorbar(aes(ymin = mean_quality - se_quality,
                    ymax = mean_quality + se_quality)) +
  geom_line() +
  geom_point() +
  theme_bw() +
  labs(x = "week",
       y = "average size of pears",
       colour = "climate")
```



```
ggplot(time.data2,aes(x = time, y = quality, colour = climate, group = pear_id)) +
  geom_line() +
  theme_bw() +
  labs(x = "week",
       y = "size of pears",
       colour = "climate")
```

```
## Warning: Removed 203 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



```
soil.data <- read.csv("D:/UHasselt/2nd semester/PMHD/project/G12.soil.data.csv",
                      sep = ",")
str(soil.data)
```

```
## 'data.frame':   96 obs. of  23 variables:
## $ tree_ID      : chr  "BE_1_1" "BE_1_2" "BE_1_3" "BE_1_4" ...
## $ location     : chr  "BE" "BE" "BE" "BE" ...
## $ ecotr_id     : int   1 1 1 1 1 1 2 2 2 2 ...
## $ tree_id      : int   1 2 3 4 5 6 7 8 9 10 ...
## $ species      : chr   "Conference" "Doyenne" "Conference" "Doyenne" ...
## $ climate      : chr   "Scenario 1" "Scenario 1" "Scenario 1" "Scenario 1" ...
## $ Bulk.Density : num   1.59 1.43 1.32 0.99 1.22 1 1.13 1.24 0.91 1.07 ...
## $ Infiltration : num   8.57 6.95 9.2 12.58 12.83 ...
## $ Soil.Porosity: num   38.5 43.2 49.1 65.2 50.1 ...
## $ Soil.Depth   : num   80.5 72.5 75.6 74 78.9 ...
## $ Water.Holding.Capacity: num   32.1 28 28.9 31 28.9 ...
## $ Soil.Nitrate : num   7.41 19.29 6.3 12.21 13.59 ...
## $ Soil.pH      : num   7.05 6.2 6.47 6.59 6.5 6.64 6.46 6.56 6.44 5.92 ...
## $ Phosphorus   : num   6.03 25.43 28.25 24.4 16.7 ...
## $ Potassium    : num  115 150 125 108 260 ...
## $ Earthworms   : num   45.7 46.6 25.6 43.1 49.1 ...
## $ Microbial.Biomass.C: num  198 220 133 141 258 ...
## $ Microbial.Biomass.N: num   11.8 19.3 13.2 10 21.6 ...
## $ Particulate.Organic.Matter: num   0.62 2.24 0.08 3.74 2.95 0.96 -0.06 0.09 3.43 2.94 ...
## $ Mineralizable.N: num   3.7 10.85 12.32 8.46 13.72 ...
## $ Soil.Enzymes : num   1.95 1.07 1.34 1.01 1.38 1.73 1.18 1.94 2.15 0.56 ...
```

```
## $ Soil.Respiration      : num  57.7 58.1 58.5 41 62 ...
## $ Total.Organic.Carbon : num  2.01 1.43 1.58 0.85 2.49 2.01 3.38 2 1.81 2.55 ...

time.data3 <- na.omit(time.data)
sapply(time.data3, function(x) sum(is.na(x)))

## pear_id tree_id ecotr_id species climate week_5 week_6 week_7
##      0      0      0      0      0      0      0      0
## week_8 week_9 week_10 week_11 week_12 week_13 week_14 week_15
##      0      0      0      0      0      0      0      0
## week_16 week_17 week_18 week_19 week_20 week_21 week_22 week_23
##      0      0      0      0      0      0      0      0
## week_24
##      0

time.data.long <- reshape(time.data3, direction = "long",
                           varying = c("week_5", "week_6", "week_7", "week_8", "week_9",
                                         "week_10", "week_11", "week_12", "week_13", "week_14",
                                         "week_15", "week_16", "week_17", "week_18", "week_19",
                                         "week_20", "week_21", "week_22", "week_23", "week_24"),
                           idvar = c("pear_id", "tree_id", "ecotr_id", "species", "climate"),
                           timevar = "time", times = 5:24, v.names = "quality")
str(time.data.long)

## 'data.frame': 3460 obs. of 7 variables:
## $ pear_id : int  2 3 4 5 6 7 9 10 11 12 ...
## $ tree_id : int  1 1 1 1 1 1 1 1 1 1 ...
## $ ecotr_id: int  1 1 1 1 1 1 1 1 1 1 ...
## $ species : chr  "Conference" "Conference" "Conference" "Conference" ...
## $ climate : chr  "Scenario 1" "Scenario 1" "Scenario 1" "Scenario 1" ...
## $ time : int  5 5 5 5 5 5 5 5 5 5 ...
## $ quality : num  4.09 3.51 3.56 4.02 4.07 ...
## - attr(*, "reshapeLong")=List of 4
## ..$ varying:List of 1
## .. ..$ quality: chr [1:20] "week_5" "week_6" "week_7" "week_8" ...
## .. ..- attr(*, "v.names")= chr "quality"
## .. ..- attr(*, "times")= int [1:20] 5 6 7 8 9 10 11 12 13 14 ...
## ..$ v.names: chr "quality"
## ..$ idvar : chr [1:5] "pear_id" "tree_id" "ecotr_id" "species" ...
## ..$ timevar: chr "time"

# write.csv(time.data.long, "D:/UHasselt/2nd semester/PMHD/project/time_data_clean.csv",
#           row.names = FALSE)
```